

Steps in Logarithmic Differentiation

1. $y = f(x)$ (Given)
2. $\ln y = \ln f(x)$ (Take natural logarithms and simplify)
3. $D_x[\ln y] = D_x[\ln f(x)]$ (Differentiate implicitly)
4. $\frac{1}{y} D_x y = D_x[\ln f(x)]$ (By Theorem (7.5))
5. $D_x y = f(x) D_x[\ln f(x)]$ (Multiply by $y = f(x)$)

Of course, to complete the solution it is necessary to differentiate $\ln f(x)$ at some stage after Step 3. If $f(x) < 0$ for some x , then Step 2 is invalid, since $\ln f(x)$ is undefined. However, in this event we could replace Step 1 by $|y| = |f(x)|$ and take natural logarithms, obtaining $\ln |y| = \ln |f(x)|$. If we now differentiate implicitly and use Theorem (7.15) we again arrive at Step 4. Thus negative values of $f(x)$ do not change the outcome and it is unnecessary to be concerned whether $f(x)$ is positive or negative. The method should not be used to find $f'(a)$ if $f(a) = 0$, since $\ln 0$ is undefined.

Example 8 Use logarithmic differentiation to find $D_x y$ if $y = \frac{(5x-4)^3}{\sqrt{2x+1}}$.

Solution As in the outline we begin by taking the natural logarithm of each side and simplifying, obtaining

$$\ln y = 3 \ln (5x - 4) - \left(\frac{1}{2}\right) \ln (2x + 1).$$

Differentiating both sides with respect to x , we obtain

$$\begin{aligned} \frac{1}{y} D_x y &= 3 \frac{1}{5x-4} (5) - \left(\frac{1}{2}\right) \frac{1}{2x+1} (2) \\ &= \frac{25x+19}{(5x-4)(2x+1)}. \end{aligned}$$

Finally, multiplying both sides of the last equation by y (that is, by $(5x-4)^3/\sqrt{2x+1}$),

$$\begin{aligned} D_x y &= \frac{25x+19}{(5x-4)(2x+1)} \cdot \frac{(5x-4)^3}{\sqrt{2x+1}} \\ &= \frac{(25x+19)(5x-4)^2}{(2x+1)^{3/2}}. \end{aligned}$$

This result may be checked by applying the quotient rule to y . ■

7.4 Exercises

Evaluate the integrals in Exercises 1–34.

- 1 $\int \frac{x}{x^2 + 1} dx$
- 2 $\int \frac{1}{8x + 3} dx$
- 3 $\int \frac{1}{7 - 5x} dx$
- 4 $\int \frac{x^3}{x^4 - 5} dx$
- 5 $\int \frac{x - 2}{x^2 - 4x + 9} dx$
- 6 $\int \frac{(2 + \ln x)^3}{x} dx$
- 7 $\int \frac{x^2}{x^3 + 1} dx$
- 8 $\int_1^2 \frac{3x}{x^2 + 4} dx$
- 9 $\int_{-2}^1 \frac{1}{2x + 7} dx$
- 10 $\int_{-1}^0 \frac{1}{4 - 5x} dx$
- 11 $\int_1^4 \frac{2}{\sqrt{x}(\sqrt{x} + 4)} dx$
- 12 $\int \frac{x - 1}{3x^2 - 6x + 2} dx$
- 13 $\int (x + e^{5x}) dx$
- 14 $\int (1 + e^{-3x}) dx$
- 15 $\int \frac{\ln x}{x} dx$
- 16 $\int \frac{1}{x(\ln x)^2} dx$
- 17 $\int_1^3 e^{-4x} dx$
- 18 $\int_0^1 e^{2x+3} dx$
- 19 $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$
- 20 $\int x e^{x^2} dx$
- 21 $\int (1 + e^{\sin x}) \cos x dx$
- 22 $\int e^{\cos^2 x} \sin 2x dx$
- 23 $\int \frac{\cos x}{\sin x} dx$
(Hint: Let $u = \sin x$.)
- 24 $\int \frac{3 \sin x}{1 + 2 \cos x} dx$
- 25 $\int e^x \cos e^x dx$
- 26 $\int \frac{\cos x - \sin x}{\cos x + \sin x} dx$
- 27 $\int \frac{(e^x + 1)^2}{e^x} dx$
- 28 $\int \frac{e^x}{(e^x + 1)^2} dx$
- 29 $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$
- 30 $\int \frac{e^x}{e^x + 1} dx$
- 31 $\int \frac{1}{x^2 + 2x + 1} dx$
- 32 $\int \frac{(x^2 - 4)^2}{2x} dx$
- 33 $\int \frac{2x^2 - 5x - 7}{x - 3} dx$
- 34 $\int \frac{x^2 + 3x + 1}{x} dx$

In Exercises 35–38 find the area of the region bounded by the graphs of the given equations.

- 35 $y = e^{2x}$, $y = 0$, $x = 0$, $x = \ln 3$
- 36 $xy = 1$, $y = 0$, $x = 1$, $x = e$
- 37 $y = e^{-x}$, $xy = 1$, $x = 1$, $x = 2$
- 38 $y = e^{-2x}$, $y = -e^x$, $x = 0$, $x = 2$
- 39 The region bounded by the graphs of $y = e^{-x^2}$, $y = 0$, $x = 0$, and $x = 1$ is revolved about the y -axis. Find the volume of the resulting solid.
- 40 The region bounded by the graphs of $y = 1/\sqrt{x}$, $y = 0$, $x = 1$, and $x = 4$ is revolved about the x -axis. Find the volume of the resulting solid.
- 41 A particle moves on a coordinate line such that its velocity at time t is e^{-3t} . If the particle is at the origin at $t = 0$, how far does it travel during the time interval $[0, 2]$?
- 42 A solid has, for its base, the region in the xy -plane bounded by the graphs of $y = e^{2x}$, $y = e^x$, and $x = 1$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is a square.

In Exercises 43 and 44 (a) set up an integral for finding the arc length L of the graph of the given equation from A to B ; (b) write out the formula for approximating L by means of the Trapezoidal Rule with $n = 5$; and (c) use a calculator to find the value of the approximation given by the formula in part (b).

- 43 $y = e^x$; $A(0, 1)$, $B(1, e)$
- 44 $y = e^{-x^2}$; $A(0, 1)$, $B(1, e^{-1})$

In Exercises 45–52 find $D_x y$ by logarithmic differentiation.

- 45 $y = (5x + 2)^3(6x + 1)^2$
- 46 $y = \sqrt{4x + 7}(x - 5)^3$
- 47 $y = (x^2 + 3)^5/\sqrt{x + 1}$
- 48 $y = (x + 1)^2(x + 2)^3(x + 3)^4$
- 49 $y = \sqrt[3]{2x + 1}(4x - 1)^2(3x + 5)^4$
- 50 $y = (2x - 3)^2/\sqrt{x + 1}(7x + 2)^3$
- 51 $y = \sqrt{(3x^2 + 2)}\sqrt{6x - 7}$
- 52 $y = (x^2 + 3)^{2/3}(3x - 4)^4/\sqrt{x}$

In Exercises 53–56 find $f'(x)$.

53 $f(x) = \ln |3 - 2x|$

54 $f(x) = \ln |4 - 5x^3|^2$

55 $f(x) = \ln |1 - e^{-2x}|^3$

56 $f(x) = 1/\ln |2x - 15|$

57 The pressure p and volume v of an expanding gas are related by the equation $pv = k$, where k is a constant. Show that the work done if the gas expands from v_0 to v_1 is $k \ln (v_1/v_0)$. (Hint: See Example 4 in Section 6.5.)

7.5 General Exponential and Logarithmic Functions

We shall now define a^x , where $a > 0$ and x is any real number. If the exponent is a *rational* number r , then applying (ii) of Theorem (7.11) and (iii) of Theorem (7.6) we see that

$$a^r = e^{\ln a^r} = e^{r \ln a}.$$

This formula is the motivation for the following definition of a^x .

Definition of a^x (7.18)

$$a^x = e^{x \ln a}$$

for every $a > 0$ and every real number x .

The function f defined by $f(x) = a^x$ is called the **exponential function with base a** . Since e^x is positive for all x , so is a^x . To approximate values of a^x , we may refer to tables of logarithmic and exponential functions, or use a calculator.

It is now possible to prove that the law of logarithms stated in (iii) of Theorem (7.6) is also true for irrational exponents. Thus, if u is any *real* number, then by Definition (7.18) and (i) of Theorem (7.11),

$$\ln a^u = \ln e^{u \ln a} = u \ln a.$$

The next theorem states that properties of rational exponents studied in elementary algebra are also true for real exponents.

Laws of Exponents (7.19)

Suppose $a > 0$ and $b > 0$. If u and v are any real numbers, then

$$\begin{aligned} a^u a^v &= a^{u+v} & (a^u)^v &= a^{uv} & (ab)^u &= a^u b^u \\ \frac{a^u}{a^v} &= a^{u-v} & \left(\frac{a}{b}\right)^u &= \frac{a^u}{b^u} \end{aligned}$$

Proof To show that $a^u a^v = a^{u+v}$ we use Definition (7.18) and (i) of Theorem (7.12) as follows:

$$\begin{aligned} a^u a^v &= e^{u \ln a} e^{v \ln a} \\ &= e^{u \ln a + v \ln a} \\ &= e^{(u+v) \ln a} \\ &= a^{u+v}. \end{aligned}$$

To prove that $(a^u)^v = a^{uv}$ we first use Definition (7.18) with a^u in place of a and $v = x$ to write

$$(a^u)^v = e^{v \ln a^u}.$$

Using the fact that $\ln a^u = u \ln a$, and then applying Definition (7.18), we obtain

$$(a^u)^v = e^{vu \ln a} = a^{vu} = a^{uv}.$$

The proofs of the remaining laws are left as exercises. □

Theorem (7.20)

- (i) $D_x(a^x) = a^x \ln a$
(ii) $D_x(a^u) = (a^u \ln a) D_x u$, where $u = g(x)$.

Proof Applying Definition (7.18) and Theorem (7.14),

$$D_x(a^x) = D_x(e^{x \ln a}) = e^{x \ln a} D_x(x \ln a) = e^{x \ln a} (\ln a).$$

Again using (7.18), we obtain

$$D_x(a^x) = a^x \ln a.$$

This gives us (i). Formula (ii) may be obtained by using the Chain Rule. □

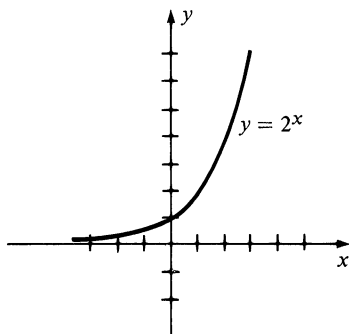
It should be noted that if $a = e$, then (i) of Theorem (7.20) reduces to (7.13). If $a > 1$, then $\ln a > 0$ and, therefore, $D_x(a^x) > 0$. Hence a^x is increasing on the interval $(-\infty, \infty)$ if $a > 1$. If $0 < a < 1$, then $\ln a < 0$ and $D_x(a^x) < 0$; that is, a^x is decreasing for all x .

The graphs of $y = 2^x$ and $y = (\frac{1}{2})^x = 2^{-x}$ illustrated in Figure 7.10 may be sketched by plotting some representative points. The graph of the equation $y = a^x$ has the general shape illustrated in (i) or (ii) of Figure 7.10, depending on whether $a > 1$ or $0 < a < 1$, respectively.

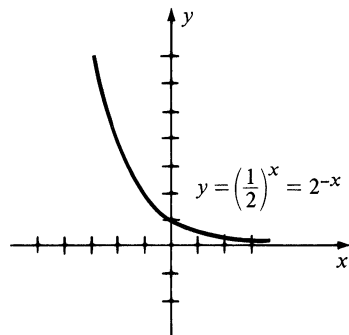
Example 1 Find y' if $y = 3^{\sqrt{x}}$.

Solution Using (ii) of Theorem (7.20) with $a = 3$ and $u = \sqrt{x}$,

$$y' = (3^{\sqrt{x}} \ln 3) \left(\frac{1}{2} x^{-1/2} \right) = \frac{3^{\sqrt{x}} \ln 3}{2\sqrt{x}}. \quad \blacksquare$$



(i)



(ii)

FIGURE 7.10

If $u = g(x)$ it is very important to distinguish between expressions of the form a^u and u^a where a is a real number. To differentiate a^u we use (7.20), whereas for u^a the Power Rule must be employed, as illustrated in the next example.

Example 2 Find y' if $y = (x^2 + 1)^{10} + 10^{x^2+1}$.

Solution Using the Power Rule for Functions and Theorem (7.20), we obtain

$$\begin{aligned} y' &= 10(x^2 + 1)^9(2x) + 10^{x^2+1} \ln 10(2x) \\ &= 20x[(x^2 + 1)^9 + 10^{x^2} \ln 10]. \end{aligned}$$

■

The integration formula in (i) of the next theorem may be verified by showing that the integrand is the derivative of the expression on the right side of the equation. Formula (ii) follows from (ii) of Theorem (7.20).

Theorem (7.21)

$$\begin{aligned} \text{(i)} \quad & \int a^x dx = \left(\frac{1}{\ln a} \right) a^x + C \\ \text{(ii)} \quad & \int a^u du = \left(\frac{1}{\ln a} \right) a^u + C, \quad \text{where } u = g(x). \end{aligned}$$

Example 3 Evaluate $\int 3^{x^2} x dx$.

Solution If we let $u = x^2$, then $du = 2x dx$ and we may proceed as follows:

$$\begin{aligned} \int 3^{x^2} x dx &= \frac{1}{2} \int 3^{x^2} (2x) dx = \frac{1}{2} \int 3^u du \\ &= \frac{1}{2} \left(\frac{1}{\ln 3} \right) 3^u + C = \left(\frac{1}{2 \ln 3} \right) 3^{x^2} + C. \end{aligned}$$

■

If $a \neq 1$ and $f(x) = a^x$, then f is a one-to-one function. Its inverse function is denoted by $\log_a$ and is called the *logarithmic function with base a* . Another way of stating this relationship is as follows.

Definition of $\log_a x$ (7.22)

$$y = \log_a x \quad \text{if and only if} \quad x = a^y.$$

The expression $\log_a x$ is called **the logarithm of x with base a** . Using this terminology, natural logarithms are logarithms with base e , that is,

$$\ln x = \log_e x.$$

The fact that laws of logarithms similar to Theorem (7.6) are true for logarithms with base a is left as an exercise.

In order to obtain the relationship between $\log_a$ and $\ln$, consider $y = \log_a x$ or, equivalently, $x = a^y$. Taking the natural logarithm of both sides of the last equation gives us $\ln x = y \ln a$, or $y = \ln x / \ln a$. This proves that

$$\log_a x = \frac{\ln x}{\ln a}.$$

Differentiating both sides of the last equation leads to (i) of the next theorem. Using the Chain Rule and generalizing to absolute values as in Theorem (7.15), gives us (ii).

Theorem (7.23)

$$(i) \quad D_x \log_a x = \frac{1}{x \ln a}.$$

$$(ii) \quad D_x \log_a |u| = \frac{1}{u \ln a} D_x u, \quad \text{where } u = g(x) \neq 0.$$

Logarithms with base 10 are useful for certain numerical computations. It is customary to refer to such logarithms as **common logarithms** and to use the symbol **log x** as an abbreviation for $\log_{10} x$. This notation is used in the next example.

Example 4 Find $f'(x)$ if $f(x) = \log \sqrt[3]{(2x + 5)^2}$.

Solution We begin by using the analogue of (iii) of Theorem (7.6) for logarithms with any base together with the fact that $(2x + 5)^2 = |2x + 5|^2$. Thus

$$f(x) = \frac{1}{3} \log |2x + 5|^2 = \frac{2}{3} \log |2x + 5|.$$

Next, employing (ii) of Theorem (7.23) with $a = 10$,

$$f'(x) = \frac{2}{3} \frac{1}{(2x + 5) \ln 10} (2) = \frac{4}{(6x + 15) \ln 10}. \quad \blacksquare$$

Since irrational exponents have been given meaning, we may now consider the **general power function** f defined by $f(x) = x^c$ where c is any real number. If c is irrational, then by definition, the domain of f is the set of positive real numbers. Using Definition (7.18), and Theorems (7.14) and (7.4),

$$\begin{aligned} D_x x^c &= D_x (e^{c \ln x}) = e^{c \ln x} D_x (c \ln x) \\ &= e^{c \ln x} \left(\frac{c}{x} \right) = x^c \left(\frac{c}{x} \right) = cx^{c-1}. \end{aligned}$$

This proves that the Power Rule is true for irrational as well as rational exponents. The Power Rule for Functions may also be extended to irrational exponents.

Example 5 Find y' if (a) $y = x^{\sqrt{2}}$; (b) $y = (1 + e^{2x})^\pi$.

Solution

$$(a) \quad y' = \sqrt{2}x^{\sqrt{2}-1}$$

$$\begin{aligned} (b) \quad y' &= \pi(1 + e^{2x})^{\pi-1} D_x(1 + e^{2x}) \\ &= \pi(1 + e^{2x})^{\pi-1} (2e^{2x}) \\ &= 2\pi e^{2x}(1 + e^{2x})^{\pi-1}. \end{aligned}$$

■

Example 6 Find $D_x y$ if $y = x^x$, where $x > 0$.

Solution Since the exponent in x^x is a variable, the Power Rule may not be used. Similarly, Theorem (7.20) is not applicable since the base a is not a fixed real number. However, by Definition (7.18), $x^x = e^{x \ln x}$ for all $x > 0$, and hence

$$\begin{aligned} D_x(x^x) &= D_x(e^{x \ln x}) \\ &= e^{x \ln x} D_x(x \ln x) \\ &= e^{x \ln x} \left[x \left(\frac{1}{x} \right) + (1) \ln x \right] \\ &= x^x(1 + \ln x). \end{aligned}$$

Another way of solving this problem is to use the method of logarithmic differentiation introduced in the previous section. In this case we take the natural logarithm of both sides of the equation $y = x^x$, and then differentiate implicitly as follows:

$$\begin{aligned} \ln y &= \ln x^x = x \ln x \\ D_x(\ln y) &= D_x(x \ln x) \\ \frac{1}{y} D_x y &= 1 + \ln x \\ D_x y &= y(1 + \ln x) = x^x(1 + \ln x). \end{aligned}$$

■

We shall conclude this section by expressing the number e as a limit.

Theorem (7.24)

$$(i) \quad \lim_{h \rightarrow 0} (1 + h)^{1/h} = e \qquad (ii) \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

Proof Applying the derivative formula (3.6) to $f(x) = \ln x$ and using laws of logarithms gives us

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \ln \frac{x+h}{x} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \ln \left(1 + \frac{h}{x}\right) \\ &= \lim_{h \rightarrow 0} \ln \left(1 + \frac{h}{x}\right)^{1/h}. \end{aligned}$$

Since $f'(x) = 1/x$ we have, for $x = 1$,

$$1 = \lim_{h \rightarrow 0} \ln(1+h)^{1/h}.$$

We next observe, from Theorem (7.11), that

$$(1+h)^{1/h} = e^{\ln(1+h)^{1/h}}.$$

Since the natural exponential function is continuous at 1, it follows from Theorem (2.27) that

$$\begin{aligned} \lim_{h \rightarrow 0} (1+h)^{1/h} &= \lim_{h \rightarrow 0} [e^{\ln(1+h)^{1/h}}] \\ &= e^{[\lim_{h \rightarrow 0} \ln(1+h)^{1/h}]} \\ &= e^1 = e. \end{aligned}$$

This establishes part (i) of the theorem. The limit in part (ii) may be obtained by introducing the change of variable $n = 1/h$ (see Exercise 52). $\square$

The formulas in Theorem (7.24) are very important in mathematics. Indeed, they are often used to *define* the number e . If a calculator is available the reader will find it instructive to calculate $(1+h)^{1/h}$ for numerically small values of h . Some approximate values are given in the following table.

h	$(1+h)^{1/h}$	h	$(1+h)^{1/h}$
0.01	2.704814	-0.01	2.731999
0.001	2.716924	-0.001	2.719642
0.0001	2.718146	-0.0001	2.718418
0.00001	2.718268	-0.00001	2.718295
0.000001	2.718280	-0.000001	2.718283

To five decimal places, $e \approx 2.71828$.

7.5 Exercises

In Exercises 1–28 find $f'(x)$ if $f(x)$ is defined by the given expression.

- | | |
|--|---|
| 1 7^x | 2 5^{-x} |
| 3 8^{x^2+1} | 4 $9^{\sqrt{x}}$ |
| 5 $\log(x^4 + 3x^2 + 1)$ | 6 $\log_3 6x - 7 $ |
| 7 5^{3x-4} | 8 3^{2-x^2} |
| 9 $(x^2 + 1)10^{1/x}$ | 10 $(10^x + 10^{-x})^{10}$ |
| 11 $7^{\sqrt{x^4+9}}$ | 12 $x/(6^x + x^6)$ |
| 13 $\log(3x^2 + 2)^5$ | 14 $\log \sqrt{x^2 + 1}$ |
| 15 $\log_5 \left \frac{6x+4}{2x-3} \right $ | 16 $\log \left \frac{1-x^2}{2-5x^3} \right $ |
| 17 $\log \ln x$ | 18 $\ln \log x$ |
| 19 $x^e + e^x$ | 20 $x^\pi \pi^x$ |
| 21 $(x+1)^x$ | 22 x^{x^2+4} |
| 23 $2^{\sin^2 x}$ | 24 $4^{\cos 3x}$ |
| 25 $\log(\sin 10x)$ | 26 $\sin(\log 10x)$ |
| 27 $x^{\sin x}$ | 28 $(\cos 2x)^x$ |

Evaluate the integrals in Exercises 29–42.

- | | |
|--------------------------------------|--|
| 29 $\int 10^{3x} dx$ | 30 $\int 5^{-2x} dx$ |
| 31 $\int x(3^{-x^2}) dx$ | 32 $\int \frac{(2^x + 1)^2}{2^x} dx$ |
| 33 $\int \frac{2^x}{2^x + 1} dx$ | 34 $\int \frac{3^x}{\sqrt{3^x + 4}} dx$ |
| 35 $\int_1^2 5^{-2x} dx$ | 36 $\int_{-1}^1 2^{3x-1} dx$ |
| 37 $\int x^2 2^{x^3} dx$ | 38 $\int \frac{10^{\sqrt{x}}}{\sqrt{x}} dx$ |
| 39 $\int (3 - 3^{\cos x}) \sin x dx$ | 40 $\int 4^x \cos 2^{2x} dx$ |
| 41 $\int \frac{1}{x \log x} dx$ | 42 $\int \frac{10^x + 10^{-x}}{10^x - 10^{-x}} dx$ |

- 43 Find the area of the region bounded by the graphs of $y = 2^x$, $x + y = 1$, and $x = 1$.

- 44 The region under the graph of $y = 3^{-x}$ from $x = 1$ to $x = 2$ is revolved about the x -axis. Find the volume of the resulting solid.
- 45 Find equations of the tangent and normal lines to the graph of the equation $y = x3^x$ at the point $P(1, 3)$.
- 46 Find the x -coordinate of the point on the graph of $y = 2^x$ at which the tangent line is parallel to the line with equation $2y - 8x + 3 = 0$.
- 47 The number of bacteria in a certain culture at time t is given by $Q(t) = 2(3^t)$, where t is measured in hours and $Q(t)$ in thousands. Show that the rate of change of $Q(t)$ at time t is proportional to $Q(t)$.
- 48 If 10 g of salt is added to a certain quantity of water, then the amount $q(t)$ that remains undissolved after t minutes is given by $q(t) = 10(4/5)^t$. Show that the rate of change of $q(t)$ at time t is proportional to $q(t)$.
- 49 Chemists use a number denoted by pH to describe quantitatively the acidity or basicity of solutions. By definition, $\text{pH} = -\log[\text{H}^+]$, where $[\text{H}^+]$ is the hydrogen ion concentration in moles per liter. For a certain brand of vinegar it is estimated (with a maximum percentage error of 0.5%) that $[\text{H}^+] \approx 6.3 \times 10^{-3}$. Calculate the pH and use differentials to estimate the maximum error in the calculation.
- 50 Using the Richter scale, the magnitude R of an earthquake of intensity I may be found by means of the formula $R = \log(I/I_0)$, where I_0 is a certain minimum intensity. Suppose it is estimated that an earthquake has an intensity that is 100 times that of I_0 . If there is a maximum percentage error of 1% in the estimate, use differentials to approximate the maximum percentage error in the calculated value of R .
- 51 The loudness of sound, as experienced by the human ear, is based upon intensity levels. A formula used for finding the intensity level α that corresponds to a sound intensity I is $\alpha = 10 \log(I/I_0)$ decibels where I_0 is a special value of I agreed to be the weakest sound that can be detected by the ear under certain conditions. Find the rate of change of α with respect to I if
- I is 10 times as great as I_0 .
 - I is 1,000 times as great as I_0 .
 - I is 10,000 times as great as I_0 . (This is the intensity level of the average voice.)
- 52 Establish (ii) of Theorem (7.24) by using the limit in part (i) and the change of variable $n = 1/h$.

In Exercises 53 and 54 sketch the graphs of the two given equations on the same coordinate plane and discuss the relationship between the graphs.

53 $y = 10^x$, $y = \log x$

54 $y = 2^x$, $y = \log_2 x$

55 Prove that for all real numbers u and v , and any $a > 0$, $b > 0$: $(ab)^u = a^u b^u$; $a^u/a^v = a^{u-v}$.

56 Prove the Laws of Logarithms:

(i) $\log_a (uv) = \log_a u + \log_a v$

(ii) $\log_a \frac{u}{v} = \log_a u - \log_a v$

(iii) $\log_a u^r = r \log_a u$

7.6 Laws of Growth and Decay

Suppose that a physical quantity varies with time and that the magnitude of the quantity at time t may be approximated by $q(t)$, where q is a differentiable function. The derivative $q'(t)$ may then be used to measure the rate of change of $q(t)$ with respect to time. In many applications, the rate of change at a given instant is directly proportional to the magnitude of the quantity at that instant. The number of bacteria in certain cultures behaves in this way. If the number of bacteria is small, then the rate of increase is small; however, as the number of bacteria increases, the rate of increase becomes larger. In many cases, this change in the rate of change is very great and we may regard the result as a microscopic view of a population explosion. The decay of a radioactive substance obeys a similar law, since as the amount of matter decreases, the rate of decay—that is, the amount of radiation—also decreases. As a final illustration, suppose an electrical condenser is allowed to discharge. If there is a large charge on the condenser at the outset, the rate of discharge is also large, but as the charge becomes weaker, the condenser discharges less rapidly. There are many other examples of this phenomenon. All of them are special cases of the general situation described in this section.

Suppose q is a differentiable function of t and $q(t) > 0$ for all t under consideration. If the rate of change of q with respect to t is directly proportional to $q(t)$, then this fact can be expressed by means of the differential equation

$$q'(t) = cq(t)$$

where c is a real number. An equivalent equation is

$$\frac{q'(t)}{q(t)} = c.$$

If the interval $[0, x]$ is in the domain of q'/q , then

$$\int_0^x \frac{q'(t)}{q(t)} dt = \int_0^x c dt.$$

Since $\ln q(t)$ is an antiderivative of $q'(t)/q(t)$, it follows from the Fundamental Theorem of Calculus that

$$\ln q(t) \Big|_0^x = ct \Big|_0^x.$$

This can also be written in any of the following equivalent forms:

$$\ln q(x) - \ln q(0) = cx$$

$$\ln \frac{q(x)}{q(0)} = cx$$

$$\frac{q(x)}{q(0)} = e^{cx}$$

$$q(x) = q(0)e^{cx}.$$

If the variable t is used instead of x , then the last equation takes on the form $q(t) = q(0)e^{ct}$. We have proved the following theorem.

Theorem (7.25)

If $q'(t) = cq(t)$, then $q(t) = q(0)e^{ct}$.

The preceding theorem states that *if the rate of change of $q(t)$ is directly proportional to $q(t)$, then q may be expressed in terms of an exponential function*. If q increases with t , the formula for $q(t)$ is called the **law of growth** of q , whereas if q decreases, it is referred to as the **law of decay** of q . The number $q(0)$ is sometimes called the **initial value** of q .

In applied problems the preceding discussion is often phrased in terms of a differential equation in which the expression $q(t)$ is suppressed. Specifically, if we let $y = q(t)$, then the differential equation $q'(t) = cq(t)$ may be written as

$$\frac{dy}{dt} = cy$$

or, in terms of differentials, as

$$dy = (cy) dt.$$

Dividing both sides of this equation by y we obtain

$$\frac{1}{y} dy = c dt.$$

Since it is possible to separate the variables y and t in the sense that they can be placed on opposite sides of the equals sign, the differential equation $dy/dt = cy$ is called a **separable differential equation**. Such equations will be studied in more detail later in the text, and it will be shown that solutions can be found by integrating both sides of the “separated” equation. Thus,

$$\int \frac{1}{y} dy = \int c dt$$

and, assuming $y > 0$,

$$\ln y = ct + d$$

where the two constants of integration have been combined into the one constant d . It follows that

$$y = e^{ct+d} = e^d e^{ct}.$$

If y_0 denotes the initial value of y (that is, the value corresponding to $t = 0$), then letting $t = 0$ in the last equation gives us

$$y_0 = e^d e^0 = e^d$$

and hence the solution $y = e^d e^{ct}$ may be written

$$y = y_0 e^{ct}.$$

This is an alternative form of the conclusion of Theorem (7.25).

Example 1 The number of bacteria in a certain culture increases from 600 to 1800 in 2 hours. Assuming that the exponential law of growth holds, find a formula for the number of bacteria in the culture at any time t . What is the number of bacteria at the end of 4 hours?

Solution Let $t = 0$ correspond to the time at which 600 bacteria are present in the culture and let $q(t)$ denote the number of bacteria after t hours. Thus $q(0) = 600$ and $q(2) = 1800$. Since the growth function is exponential, $q(t)$ has the form stated in Theorem (7.25) with $q(0) = 600$; that is,

$$q(t) = 600e^{ct}.$$

Letting $t = 2$ we obtain

$$1800 = 600e^{2c}, \quad \text{or} \quad e^{2c} = 3.$$

Consequently,

$$2c = \ln 3, \quad \text{or} \quad c = \frac{1}{2} \ln 3.$$

Thus the formula for $q(t)$ is

$$q(t) = 600e^{[(1/2)\ln 3]t}.$$

If we use the fact that $e^{\ln 3} = 3$ (see (Definition (7.10))), then this law of growth can be expressed in terms of an exponential function with base 3 as follows:

$$q(t) = 600(e^{\ln 3})^{t/2} = 600(3)^{t/2}.$$

In particular, at the end of 4 hours,

$$q(4) = 600(3)^{4/2} = 600(9) = 5400. \quad \blacksquare$$

Example 2 Radium decays exponentially and has a half-life of approximately 1600 years; that is, given any quantity, one-half of it will disintegrate in 1600 years. Find a formula for the amount $q(t)$ remaining from 50 mg of pure radium after t years. When will there be 20 mg left?

Solution By hypothesis, $q(0) = 50$ and $q(1600) = 25$. Applying Theorem (7.25),

$$q(t) = 50e^{ct}.$$

Letting $t = 1600$ we obtain

$$25 = 50e^{1600c}, \quad \text{or} \quad e^{1600c} = \frac{1}{2}.$$

Hence $1600c = \ln \frac{1}{2} = \ln 1 - \ln 2 = -\ln 2$

and $c = -\frac{\ln 2}{1600}.$

Consequently, $q(t) = 50e^{-(\ln 2/1600)t}.$

As in Example 1 we may write this formula in terms of a different base as follows:

$$q(t) = 50(e^{\ln 2})^{-t/1600}$$

or $q(t) = 50(2)^{-t/1600}.$

To find the value of t at which $q(t) = 20$ it is necessary to solve the equation

$$20 = 50(2)^{-t/1600} \quad \text{or} \quad 2^{t/1600} = \frac{5}{2}.$$

Taking the natural logarithm of both sides,

$$\frac{t}{1600} \ln 2 = \ln \frac{5}{2}, \quad \text{or} \quad t = \frac{1600 \ln \frac{5}{2}}{\ln 2}.$$

If an approximation is desired, then using a calculator or Table C,

$$t \approx 1600(0.916)/(0.693) \approx 2,115 \text{ years.} \quad \blacksquare$$

Example 3 According to Newton's Law of Cooling, the rate at which an object cools is directly proportional to the difference in temperature between the object and the surrounding medium. If a certain object cools from 125° to 100° in half an hour when surrounded by air at a temperature of 75° , find its temperature at the end of another half hour.

Solution If y denotes the temperature of the object after t hours of cooling, then by Newton's Law,

$$\frac{dy}{dt} = c(y - 75)$$

where c is a constant. Separating variables we obtain

$$\frac{1}{y - 75} dy = c dt.$$

Integrating both sides leads to

$$\ln(y - 75) = ct + b$$

where b is a constant or, equivalently,

$$y - 75 = e^{ct+b} = e^b e^{ct}.$$

If we let $k = e^b$, then the last formula may be written

$$y = ke^{ct} + 75.$$

Since $y = 125$ when $t = 0$ we see that

$$125 = ke^0 + 75 = k + 75 \quad \text{or} \quad k = 50.$$

Hence

$$y = 50e^{ct} + 75.$$

Since $y = 100$ when $t = \frac{1}{2}$ hour,

$$100 = 50e^{c/2} + 75 \quad \text{or} \quad e^{c/2} = \frac{25}{50} = \frac{1}{2}.$$

This implies that

$$c = 2 \ln \frac{1}{2} = \ln \frac{1}{4}.$$

Consequently, the temperature y after t hours is given by

$$y = 50e^{t \ln(1/4)} + 75.$$

In particular, if $t = 1$,

$$\begin{aligned} y &= 50e^{\ln(1/4)} + 75 \\ &= 50\left(\frac{1}{4}\right) + 75 = 87.5^\circ. \end{aligned}$$

■

In many cases, growth circumstances are more stable than that described in Theorem (7.25). Typical situations that occur involve populations, sales of products, and values of assets. In biology, a function G is sometimes used as follows to estimate the size of a quantity at time t :

$$G(t) = ke^{(-Ae^{-Bt})}$$

where k , A , and B are positive constants. The function G is called a **Gompertz growth function**. It is always positive and increasing, but has a limit as t increases without bound. The graph of G is called a **Gompertz growth curve**.

Example 4 Discuss and sketch the graph of the Gompertz growth function G .

Solution We first observe that the y -intercept is $G(0) = ke^{-A}$ and that $G(t) > 0$ for all t . Differentiating, we obtain

$$\begin{aligned} G'(t) &= ke^{(-Ae^{-Bt})} D_t(-Ae^{-Bt}) \\ &= ABke^{(-Bt - Ae^{-Bt})} \end{aligned}$$

and

$$\begin{aligned} G''(t) &= ABke^{(-Bt - Ae^{-Bt})} D_t(-Bt - Ae^{-Bt}) \\ &= ABk(-B + AB e^{-Bt}) e^{-Bt - Ae^{-Bt}}. \end{aligned}$$

Since $G'(t) > 0$ for all t , the function G is increasing on $[0, \infty)$. The second derivative $G''(t)$ is zero if

$$-B + AB e^{-Bt} = 0, \quad \text{or} \quad e^{Bt} = A.$$

Solving the last equation for t gives us $t = (1/B) \ln A$, which is a critical number for the function G' . It is left as an exercise to show that at this time the rate of growth G' has a maximum value Bk/e . It is also left to the reader to show that

$$\lim_{t \rightarrow \infty} G'(t) = 0 \quad \text{and} \quad \lim_{t \rightarrow \infty} G(t) = k.$$

Hence, as t increases without bound, the rate of growth approaches 0 and the graph of G has a horizontal asymptote $y = k$. A typical graph is sketched in Figure 7.11. ■

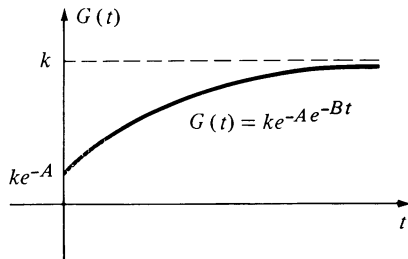


FIGURE 7.11

7.6 Exercises

- The number of bacteria in a certain culture increases from 5,000 to 15,000 in 10 hours. Assuming that the rate of increase is proportional to the number of bacteria present, find a formula for the number of bacteria in the culture at any time t . Estimate the number at the end of 20 hours. When will the number be 50,000?
- The polonium isotope ^{210}Po has a half-life of approximately 140 days. If a sample weighs 20 mg initially, how much remains after t days? Approximately how much will be left after two weeks?
- If the temperature is constant, then the rate of change of barometric pressure p with respect to altitude h is proportional to p . If $p = 30$ in. at sea level and $p = 29$ in. when $h = 1000$ ft, find the pressure at an altitude of 5000 ft.
- The population of a certain city is increasing at the rate of 5 percent per year. If the present population is 500,000, what will the population be in 10 years?
- It is usually assumed that $\frac{1}{4}$ acre of land is required to provide food for one person. It is also estimated that there are 10 billion acres of tillable land in the world, and hence a maximum population of 40 billion people can be sustained if no other food source is available. The world population at the beginning of 1980 was approximately 4.5 billion. Assuming that the population increases at a rate of 2 percent per year, when will the maximum population be reached?
- A metal plate that has been heated cools from 180° to 150° in 20 minutes when surrounded by air at a temperature of 60° . Use Newton's Law of Cooling (see Example 3) to approximate its temperature at the end of one hour of cooling. When will the temperature be 100° ?
- An outdoor thermometer registers a temperature of 40° . Five minutes after it is brought into a room where the temperature is 70° , the thermometer registers 60° . When will it register 65° ? (See Example 3.)
- The rate at which salt dissolves in water is directly proportional to the amount that remains undissolved. If 10 lb of salt is placed in a container of water and 4 lb dissolves in 20 minutes, how long will it take 2 more pounds to dissolve?
- According to one of Kirchhoff's rules for electrical circuits, $E = Ri + L(di/dt)$ where the constants E , R , and L denote the electromotive force, the resistance, and the inductance, respectively, and i denotes the current at time t (see figure). If the electromagnetic force is terminated at time $t = 0$ and if the current is I at the instant of removal, prove that $i = Ie^{-Rt/L}$.

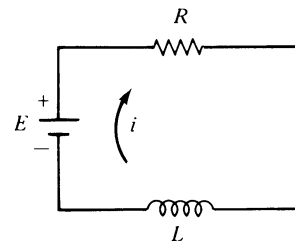


FIGURE FOR EXERCISE 9

- 10 The rate at which sugar decomposes in solution is directly proportional to the amount of sugar remaining. If a solution contains k pounds of sugar initially and if y is the amount which decomposes in time t , express y in terms of t .
- 11 If a sum of money P is invested at an interest rate of 100r percent per year, compounded m times per year, then the principal at the end of t years is given by $P(1 + rm^{-1})^{mt}$. If we regard m as a real number and let m increase without bound, then the interest is said to be *compounded continuously*. Use Theorem (7.24) to show that in this case the principal after t years is Pe^{rt} .
- 12 Use the results of Exercise 11 to find the principal after one year if \$1000 is invested at a rate of 10 percent, compounded (a) monthly; (b) daily; (c) hourly; (d) continuously.
- 13 In Example 4, verify that (a) Bk/e is a maximum value for G' ; (b) $\lim_{t \rightarrow \infty} G'(t) = 0$; (c) $\lim_{t \rightarrow \infty} G(t) = k$.
- 14 Sketch the graph of G in Example 4 if $k = 10$, $A = 1/2$, and $B = 1$.
- 15 In the **Law of Logistic Growth**, it is assumed that at time t , the rate of growth $f'(t)$ of a quantity $f(t)$ is given by $f'(t) = Af(t)[B - f(t)]$ where A and B are constants. If $f(0) = C$, show that

$$f(t) = \frac{BC}{C + (B - C)e^{-ABt}}.$$

- 16 The graph of $f(x) = a + b(1 - e^{-cx})$ where a , b , and c are positive constants may be used to describe certain learning processes. To illustrate, suppose a manufacturer estimates that a new employee can produce five items the first day on the job. As the employee becomes more proficient, more items per day can be produced until a certain maximum production is reached. Suppose that on the n th day on the job, the number $f(n)$ of items produced is approximated by $f(n) = 3 + 20(1 - e^{-0.1n})$.
- (a) Estimate the number of items produced on the fifth day; the ninth day; the twenty-fourth day; the thirtieth day.
- (b) Sketch the graph of f from $n = 0$ to $n = 30$. (Graphs of this type are called **learning curves**, and are used frequently in education and psychology.)
- (c) What happens as n increases without bound?
- 17 If a radioactive substance decays according to the law $q(t) = q_0 e^{-ct}$, where $q_0 = q(0)$, find its half-life if (a) $c = 100$; (b) $c = 1000$; (c) c is any positive constant.
- 18 A physicist finds that an unknown radioactive substance registers 2000 counts per minute on a Geiger counter. Ten days later it registers 1500 counts per minute. Approximate the half-life of the substance.
- 19 A machine purchased for \$20,000 had a value of \$16,000 after two years of use. Assuming that the value decreases exponentially, approximate the value of the machine at the end of another year.
- 20 The cost of real estate in a certain city has appreciated at a rate of 10 percent per year since 1975. If this rate continues, approximately what will a house that was purchased for \$60,000 in 1980 be worth in 1990?
- 21 Veterinarians use sodium pentobarbital to anesthetize animals. Suppose that to anesthetize a dog, 30 mg are required for each kilogram of body weight. If sodium pentobarbital is eliminated exponentially from the bloodstream, and half is eliminated in four hours, approximate the single dose that will anesthetize a dog that weighs 20 kg for 45 minutes.
- 22 In **seasonal population growth** the population $q(t)$ at time t increases during spring and summer, but decreases during the fall and winter. A mathematical formula that is sometimes used to describe this growth is $q'(t)/q(t) = k \sin 2\pi t$, where t is in years and $t = 0$ corresponds to the first day of spring. (a) Show that the population $q(t)$ is seasonal. (b) If $q_0 = q(0)$, find an explicit formula for $q(t)$.
- 23 The technique of **carbon dating** is used to determine the age of archeological or geological specimens. This method is based on the fact that the unstable carbon isotope C^{14} is present in the CO_2 in the atmosphere. Plants take in carbon from the atmosphere; when they die the C^{14} that has accumulated begins to decay, with a half-life of approximately 5700 years. By measuring the amount of C^{14} that remains in a specimen, it is possible to approximate when the organism died. Suppose that a bone which is discovered contains 20 percent as much C^{14} as an equal amount of carbon in present day bone. Approximate the age of the bone.
- 24 Refer to Exercise 23. The hydrogen isotope 3_1H , which has a half life of 12.3 years, is produced in the atmosphere by cosmic rays and is brought to earth by rain. If the wood siding of an old house contains 10 percent as much 3_1H as the siding on a similar new house, approximate its age.

7.7 Derivatives of Inverse Functions

When inverse functions were defined in Section 7.1, we introduced techniques for finding $f^{-1}(x)$ if $f(x)$ is stated in terms of a simple algebraic expression. It was also pointed out that the graphs of f and f^{-1} are reflections of one another through the line $y = x$. In Definition (7.8) we proved that the natural logarithmic and exponential functions are inverse functions of one another. Inverses of trigonometric functions will be discussed in Chapter 8. In this section we shall prove several general theorems that hold for all inverse functions, and we will obtain a formula for finding their derivatives.

Theorem (7.26)

If a function f is continuous and increasing on an interval $[a, b]$, then f has an inverse function f^{-1} that is continuous and increasing on the interval $[f(a), f(b)]$.

Proof The theorem is intuitively evident if we regard the graph of f^{-1} as a reflection, through the line $y = x$, of the graph of f . (See, for example, Figure 7.2.) However, since a graphical illustration does not constitute a proof, we shall proceed as follows.

To establish the existence of f^{-1} , it is sufficient to show that f is a one-to-one function with range $[f(a), f(b)]$. Consider numbers x_1 and x_2 in $[a, b]$. If $x_1 < x_2$, then, since f is increasing, $f(x_1) < f(x_2)$. Similarly if $x_2 < x_1$, then $f(x_2) < f(x_1)$. Consequently, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$, that is, f is one-to-one. By the Intermediate Value Theorem (2.29), f takes on every value between $f(a)$ and $f(b)$ and, therefore, the range of f is $[f(a), f(b)]$. Thus the inverse function f^{-1} exists, with domain $[f(a), f(b)]$ and range $[a, b]$.

To prove that f^{-1} is increasing, it must be shown that if $w_1 < w_2$ in $[f(a), f(b)]$, then $f^{-1}(w_1) < f^{-1}(w_2)$ in $[a, b]$. We shall give an indirect proof of this fact. Thus, suppose $f^{-1}(w_2) \leq f^{-1}(w_1)$. Since f is increasing, $f(f^{-1}(w_2)) \leq f(f^{-1}(w_1))$ and hence $w_2 \leq w_1$, which is a contradiction. Consequently $f^{-1}(w_1) < f^{-1}(w_2)$.

It remains to prove that f^{-1} is continuous on $[f(a), f(b)]$. Since f^{-1} is the inverse of f , it follows that $y = f(x)$ if and only if $x = f^{-1}(y)$. In particular, if y_0 is in the open interval $(f(a), f(b))$, let x_0 denote the number in the interval (a, b) such that $y_0 = f(x_0)$ or, equivalently, $x_0 = f^{-1}(y_0)$. We wish to show that

$$(*) \quad \lim_{y \rightarrow y_0} f^{-1}(y) = f^{-1}(y_0) = x_0.$$

It is enlightening to use a geometric representation of a function and its inverse as described in Section 7.1. In this event, the domain $[a, b]$ of f is represented by points on an x -axis and the domain $[f(a), f(b)]$ by points on a y -axis. Arrows are drawn from one axis to the other to represent functional values. To prove (*), consider any interval $(x_0 - \varepsilon, x_0 + \varepsilon)$ where $\varepsilon > 0$. It is sufficient to find an interval $(y_0 - \delta, y_0 + \delta)$, of the type sketched in (i) of

Figure 7.12, such that whenever y is in $(y_0 - \delta, y_0 + \delta)$, then $f^{-1}(y)$ is in $(x_0 - \varepsilon, x_0 + \varepsilon)$. We may assume, without loss of generality, that $x_0 - \varepsilon$ and $x_0 + \varepsilon$ are in $[a, b]$. As illustrated in (ii) of Figure 7.12, let $\delta_1 = y_0 - f(x_0 - \varepsilon)$ and $\delta_2 = f(x_0 + \varepsilon) - y_0$. Since the function f determines a one-to-one correspondence between the numbers in the intervals $(x_0 - \varepsilon, x_0 + \varepsilon)$ and $(y_0 - \delta_1, y_0 + \delta_2)$, f^{-1} maps the numbers in $(y_0 - \delta_1, y_0 + \delta_2)$ onto the numbers in $(x_0 - \varepsilon, x_0 + \varepsilon)$. Consequently, if δ denotes the smaller of δ_1 and δ_2 , then whenever y is in $(y_0 - \delta, y_0 + \delta)$, $f^{-1}(y)$ is in $(x_0 - \varepsilon, x_0 + \varepsilon)$. This proves that f^{-1} is continuous on the open interval $(f(a), f(b))$. The continuity at the endpoints $f(a)$ and $f(b)$ is proved in a similar manner using one-sided limits.

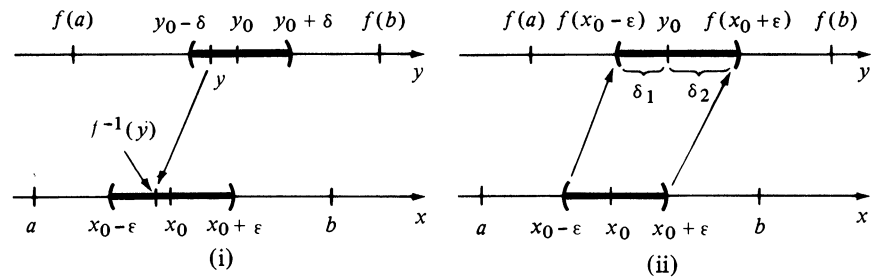


FIGURE 7.12

□

The proof of the next result is omitted since it is similar to that of Theorem (7.26).

Theorem (7.27)

If a function f is continuous and decreasing on an interval $[a, b]$, then f has an inverse function that is continuous and decreasing on the interval $[f(b), f(a)]$.

Example 1 Verify Theorem (7.26) if $f(x) = x^2 - 3$ on the interval $[0, 6]$.

Solution Since f is a polynomial function, it is continuous on $[0, 6]$. Moreover, since $f'(x) = 2x$, $f'(x) > 0$ for all x in $(0, 6]$ and, therefore, f is increasing on $[0, 6]$. It was shown in Example 2 of Section 7.1 that f^{-1} is given by

$$f^{-1}(x) = \sqrt{x + 3}, \quad \text{where } x \geq -3.$$

In the present example the domain of f^{-1} is $[f(0), f(6)]$, that is, $[-3, 33]$. Since $\lim_{x \rightarrow a} \sqrt{x + 3} = \sqrt{a + 3}$ for every a in $(-3, 33]$, f^{-1} is continuous on this interval. It is also continuous at -3 . Finally, since the derivative

$$D_x f^{-1}(x) = \frac{1}{2\sqrt{x + 3}}$$

is positive on $(-3, 33]$, it follows that f^{-1} is increasing on $[-3, 33]$. ■

The next theorem provides a method for finding the derivative of an inverse function.

Theorem (7.28)

If a differentiable function f has an inverse function g , and if $f'(g(c)) \neq 0$, then g is differentiable at c and

$$g'(c) = \frac{1}{f'(g(c))}.$$

Proof Using the formula for the derivative given in Theorem (3.7),

$$g'(c) = \lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c}.$$

Let us introduce a new variable z such that $z = g(x)$ and let $a = g(c)$. Since f and g are inverse functions of one another, it follows that

$$g(x) = z \quad \text{if and only if} \quad f(z) = x,$$

$$g(c) = a \quad \text{if and only if} \quad f(a) = c.$$

The statement $x \rightarrow c$ in the limit formula for $g'(c)$ may be replaced by $z \rightarrow a$, since if $x \rightarrow c$, then $g(x) \rightarrow g(c)$, and if $z \rightarrow a$, then $f(z) \rightarrow f(a)$. Thus we may write

$$\begin{aligned} g'(c) &= \lim_{z \rightarrow a} \frac{z - a}{f(z) - f(a)} \\ &= \lim_{z \rightarrow a} \frac{1}{\frac{f(z) - f(a)}{z - a}} \\ &= \frac{1}{f'(a)} = \frac{1}{f'(g(c))}. \end{aligned}$$

□

It is convenient to restate Theorem (7.28) as follows.

Theorem (7.29)

If g is the inverse function of a differentiable function f , and if $f'(g(x)) \neq 0$, then

$$g'(x) = \frac{1}{f'(g(x))}.$$

Example 2 Use Theorem (7.29) and the fact that the natural exponential function is the inverse of the natural logarithmic function to prove that $D_x e^x = e^x$.

Solution If $f(x) = \ln x$ and $g(x) = e^x$, then $f'(x) = 1/x$ and $f'(g(x)) = f'(e^x) = 1/e^x$. Hence, from Theorem (7.29),

$$g'(x) = \frac{1}{(1/e^x)} = e^x. \quad \blacksquare$$

Example 3 If $f(x) = x^3 + 2x - 1$, prove that f has an inverse function g and find the slope of the tangent line to the graph of g at the point $P(2, 1)$.

Solution Since f is continuous and $f'(x) = 3x^2 + 2 > 0$ for all x , we may conclude from Theorem (7.28) that f has an inverse function g . Since $f(1) = 2$ it follows that $g(2) = 1$ and consequently the point $P(2, 1)$ is on the graph of g . Using Theorem (7.29), the slope of the desired tangent line is

$$g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(1)} = \frac{1}{5}. \quad \blacksquare$$

There is an easy way to remember Theorem (7.29) if f is defined by an equation of the form $y = f(x)$. If g is the inverse function of f , then $g(y) = g(f(x)) = x$. From (7.29), we obtain

$$g'(y) = \frac{1}{f'(g(y))} = \frac{1}{f'(x)}$$

or, using differential notation,

$$\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx}\right)}.$$

This shows that in a sense, the derivative of the inverse function g is the reciprocal of the derivative of f . A disadvantage of using the last two formulas is that neither is stated in terms of the independent variable for the inverse function. To illustrate, suppose in Example 3 we let $y = x^3 + 2x - 1$ and $x = g(y)$. Then

$$\frac{dx}{dy} = \frac{1}{(dy/dx)} = \frac{1}{3x^2 + 2},$$

that is,
$$g'(y) = \frac{1}{3x^2 + 2} = \frac{1}{3(g(y))^2 + 2}.$$

This may also be written in the form

$$g'(x) = \frac{1}{3(g(x))^2 + 2}.$$

Consequently, to find $g'(x)$ it is necessary to know $g(x)$, just as in Theorem (7.29).

7.7 Exercises

In Exercises 1–10 prove that the function f , defined on the given interval, has an inverse function f^{-1} and state its domain. Find $f^{-1}(x)$. Sketch the graphs of f and f^{-1} on the same coordinate plane. Find $D_x f^{-1}(x)$ directly and also by means of Theorem (7.29).

- 1 $f(x) = \sqrt{2x + 3}$, $[1, 11]$
- 2 $f(x) = \sqrt[3]{5x + 2}$, $[0, 5]$
- 3 $f(x) = 4 - x^2$, $[0, 7]$
- 4 $f(x) = x^2 - 4x + 5$, $[-1, 1]$
- 5 $f(x) = 1/x$, $(0, \infty)$
- 6 $f(x) = \sqrt{9 - x^2}$, $[0, 3]$
- 7 $f(x) = e^{-x^2}$, $[0, \infty)$
- 8 $f(x) = \ln(3 - 2x)$, $(-\infty, \frac{3}{2})$
- 9 $f(x) = e^x - e^{-x}$, $(-\infty, \infty)$
- 10 $f(x) = e^x + e^{-x}$, $[0, \infty)$

In Exercises 11–14 prove that f has an inverse function f^{-1} on every closed interval $[a, b]$ and find the slope of the tangent line to the graph of f^{-1} at the indicated point P .

- 11 $f(x) = x^5 + 3x^3 + 2x - 1$, $P(5, 1)$
- 12 $f(x) = 2 - x - x^3$, $P(-8, 2)$
- 13 $f(x) = (e^{2x} - 1)/(e^{2x} + 1)$, $P(0, 0)$
- 14 $f(x) = e^{2-3x}$, $P(1/e, 1)$

In Exercises 15–18, prove that if the domain of f is $\mathbb{R}$, then f has no inverse function. Also prove that if the domain is suitably restricted, then f^{-1} exists.

- 15 $f(x) = x^4 + 3x^2 + 7$
- 16 $f(x) = |x - 2|$
- 17 $f(x) = 10^{-x^2}$
- 18 $f(x) = \ln(x^2 + 1)$
- 19 Complete the proof of Theorem (7.26) by showing that f^{-1} is continuous at $f(a)$ and $f(b)$.
- 20 Prove Theorem (7.27).

7.8 Review: Concepts

Define or discuss each of the following.

- 1 Inverse functions
- 2 The natural logarithmic function
- 3 The natural exponential function
- 4 Laws of logarithms
- 5 Derivative formulas for $\ln u$ and e^u
- 6 Logarithmic differentiation

- 7 a^x , where $a > 0$
- 8 The number e
- 9 $\log_a$
- 10 Derivative formulas for $\log_a u$ and a^u
- 11 General power function
- 12 Laws of growth and decay
- 13 Derivatives of inverse functions

Review: Exercises

Find $f'(x)$ if $f(x)$ is defined as in Exercises 1–24.

- 1 $(1 - 2x) \ln |1 - 2x|$
- 2 $\sqrt{\ln \sqrt{x}}$
- 3 $\ln \frac{(3x + 2)^4 \sqrt{6x - 5}}{8x - 7}$
- 4 $\log \left| \frac{2 - 9x}{1 - x^2} \right|$
- 5 $\frac{1}{\ln(2x^2 + 3)}$
- 6 $\frac{\ln x}{e^{2x} + 1}$
- 7 $e^{\ln(x^2 + 1)}$
- 8 $\ln(e^{4x} + 9)$

- 9 $10^x \log x$
- 10 $\ln \sqrt[4]{x/(3x + 5)}$
- 11 $x^{\ln x}$
- 12 $4^{\sqrt{2x+3}}$
- 13 $x^2 e^{1-x^2}$
- 14 $\sqrt{e^{3x} + e^{-3x}}$
- 15 $2^{-1/x}/(x^3 + 4)$
- 16 $5^{3x} + (3x)^5$
- 17 $(1 + \sqrt{x})^e$
- 18 $\ln e^{\sqrt{x}}$
- 19 $10^{\ln x}$
- 20 $(x^2 + 1)^{2x}$

21 $\ln \cos^4 3x$

22 $7^{\sin 3x}$

23 $(\sin x)^{\cos x}$

24 $\frac{1}{\sin^2 e^{-3x}}$

Find y' in Exercises 25 and 26.

25 $1 + xy = e^{xy}$

26 $\ln(x + y) + x^2 - 2y^3 = 1$

27 Show, by substitution, that $y = c_1 e^{ax} + c_2 x e^{ax}$ is a solution of the differential equation $y'' - 2ay' + a^2 y = 0$, where c_1 , c_2 , and a are any constants.

28 The position of a point on a coordinate line during the time interval $[-5, 5]$ is given by $f(t) = (t^2 + 2t)e^{-t}$.

- When is the velocity zero?
- When is the acceleration zero?
- When does the point move in the positive direction?
- When does the point move in the negative direction?

Evaluate the integrals in Exercises 29–46.

29 $\int \frac{(1 + e^x)^2}{e^{2x}} dx$

30 $\int_0^1 e^{-3x+2} dx$

31 $\int_1^4 \frac{1}{\sqrt{x} e^{\sqrt{x}}} dx$

32 $\int \frac{1}{x \ln x} dx$

33 $\int \frac{\cos 2x}{1 - 2 \sin 2x} dx$

34 $\int_1^2 \frac{x^2 + 1}{x^3 + 3x} dx$

35 $\int \frac{x^2}{3x + 2} dx$

36 $\int \frac{e^{1/x}}{x^2} dx$

37 $\int_0^1 x 4^{x^2} dx$

38 $\int \frac{(e^{2x} + e^{3x})^2}{e^{5x}} dx$

39 $\int \frac{1}{x \sqrt{\log x}} dx$

40 $\int \frac{x}{x^4 + 2x^2 + 1} dx$

41 $\int \frac{x^2 + 1}{x + 1} dx$

42 $\int \frac{2e^x}{1 + e^x} dx$

43 $\int 5^x e^x dx$

44 $\int x 10^{x^2} dx$

45 $\int x^e dx$

46 $\int 3^x (3 + \sin 3^x) dx$

47 A particle moves on a coordinate line with an acceleration at time t of $e^{t/2}$ cm/sec². At $t = 0$ the particle is at the origin and its velocity is 6 cm/sec. How far does it travel during the time interval $[0, 4]$?

48 Find the local extrema for $f(x) = x^2 \ln x$, $x > 0$. Discuss concavity, find the points of inflection, and sketch the graph of f .

49 Find an equation of the tangent line to the graph of $y = x e^{1/x^3} + \ln|2 - x^2|$ at the point $P(1, e)$.

50 Find the area of the region bounded by the graphs of the equations $y = e^{2x}$, $y = x/(x^2 + 1)$, $x = 0$, and $x = 1$.

51 The region bounded by the graphs of $y = e^{4x}$, $x = -2$, $x = -3$, and $y = 0$ is revolved about the x -axis. Find the volume of the resulting solid.

52 If $f(x) = 2x^3 - 8x + 5$ on $[-1, 1]$, prove that f has an inverse function g , and find $g'(5)$.

53 Suppose $f(x) = e^{2x} + 2e^x + 1$, where $x \geq 0$.

(a) Prove that f has an inverse function f^{-1} and state its domain.

(b) Find $f^{-1}(x)$ and $D_x f^{-1}(x)$.

(c) Find the slope of the tangent line to the graph of f at the point $(0, 4)$ and the slope of the tangent line to the graph of f^{-1} at $(4, 0)$.

54 A certain radioactive substance has a half-life of five days. How long will it take for an amount A to disintegrate to the time at which only 1 percent of A remains?

55 The rate at which sugar decomposes in water is proportional to the amount that remains undecomposed. Suppose that 10 lb of sugar is placed in a container of water at 1:00 P.M., and one-half is dissolved at 4:00 P.M.

(a) How long will it take two more pounds to dissolve?

(b) How much of the 10 lb will be dissolved at 8:00 P.M.?

56 According to Newton's Law of Cooling, the rate at which an object cools is directly proportional to the difference in temperature between the object and its surrounding medium. If $f(t)$ denotes the temperature at time t , show that $f(t) = T + [f(0) - T]e^{-kt}$ where T is the temperature of the surrounding medium and k is a positive constant.

57 The bacteria *E. coli* undergoes cell division approximately every 20 minutes. Starting with 100,000 cells, determine the number of cells after two hours.

8 Other Transcendental Functions

Formulas for derivatives and integrals of the sine and cosine functions were obtained in earlier chapters. We shall now consider the remaining trigonometric

functions and the inverse trigonometric functions. The last two sections contain a discussion of hyperbolic and inverse hyperbolic functions.

8.1 Derivatives of the Trigonometric Functions

Let us begin by listing formulas for the derivatives of all six trigonometric functions. In the statement of the theorem it is assumed that $u = g(x)$, where g is differentiable, and that x is restricted to values for which the indicated functions are defined.

Theorem (8.1)

$$D_x \sin u = \cos u D_x u$$

$$D_x \csc u = -\csc u \cot u D_x u$$

$$D_x \cos u = -\sin u D_x u$$

$$D_x \sec u = \sec u \tan u D_x u$$

$$D_x \tan u = \sec^2 u D_x u$$

$$D_x \cot u = -\csc^2 u D_x u$$

Proof The derivative formulas for the sine and cosine were proved in Chapter 3. To obtain the cotangent formula it is sufficient to show that $D_x \cot x = -\csc^2 x$, since $D_x \cot u$ may then be found by using the Chain Rule.

Applying the Quotient Rule to $\cot x = \cos x / \sin x$,

$$\begin{aligned} D_x \cot x &= \frac{\sin x D_x \cos x - \cos x D_x \sin x}{\sin^2 x} \\ &= \frac{\sin x(-\sin x) - \cos x(\cos x)}{\sin^2 x} \\ &= -\frac{\sin^2 x + \cos^2 x}{\sin^2 x} \\ &= -\frac{1}{\sin^2 x} = -\csc^2 x. \end{aligned}$$

Similarly, for the secant function,

$$\begin{aligned} D_x \sec x &= D_x \frac{1}{\cos x} = \frac{(\cos x) D_x 1 - (1) D_x \cos x}{\cos^2 x} \\ &= \frac{0 + \sin x}{\cos^2 x} = \frac{1}{\cos x} \frac{\sin x}{\cos x} \\ &= \sec x \tan x. \end{aligned}$$

Proofs for $\tan x$ and $\csc x$ are left as exercises. □

Example 1 Find y' if (a) $y = \cot x^4$; (b) $y = \cot^4 x$.

Solution

(a) By Theorem (8.1),

$$y' = -\csc^2 x^4 D_x(x^4) = -4x^3 \csc^2 x^4.$$

(b) First applying the Power Rule to $y = (\cot x)^4$, and then using Theorem (8.1),

$$y' = 4 \cot^3 x D_x \cot x = -4 \cot^3 x \csc^2 x. \quad \blacksquare$$

Example 2 Find $f'(x)$ if (a) $f(x) = e^{-2x} \tan 4x$; (b) $f(x) = \sec^2(3x - 1)$.

Solution

(a) First using the Product Rule and then Theorems (8.1) and (7.14),

$$\begin{aligned} f'(x) &= e^{-2x} D_x \tan 4x + (\tan 4x) D_x e^{-2x} \\ &= e^{-2x} (\sec^2 4x)(4) + (\tan 4x)(e^{-2x})(-2) \\ &= 2e^{-2x} (2 \sec^2 4x - \tan 4x). \end{aligned}$$

(b) By the Power Rule and Theorem (8.1),

$$\begin{aligned} f'(x) &= 2 \sec(3x - 1) D_x \sec(3x - 1) \\ &= 2 \sec(3x - 1) \sec(3x - 1) \tan(3x - 1)(3) \\ &= 6 \sec^2(3x - 1) \tan(3x - 1). \quad \blacksquare \end{aligned}$$

The graphs of $y = \sin x$ and $y = \cos x$ are sketched in Figure 3.9. Portions of the graphs of the other four trigonometric functions are shown in Figure 8.1. The dashed lines indicate vertical asymptotes. The reader should verify each graph by plotting several points and investigating symmetry, periodicity, maxima, minima and concavity.

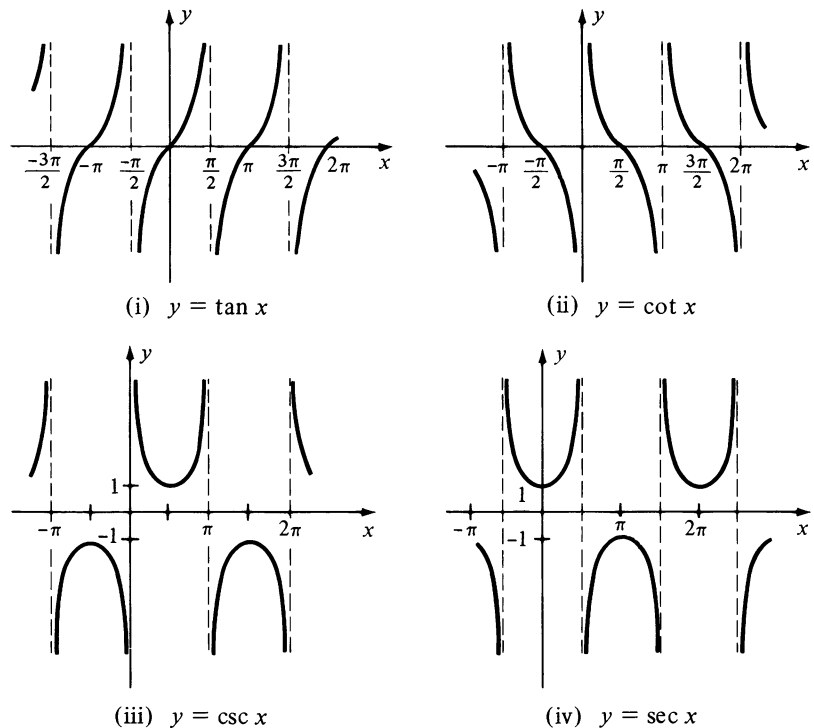


FIGURE 8.1

Example 3 Suppose $f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

Prove that f is continuous at every real number and sketch the graph of f .

Solution Let us show that $\lim_{x \rightarrow a} f(x) = f(a)$ for every real number a . If $a \neq 0$, then

$$\begin{aligned} \lim_{x \rightarrow a} \left(x \sin \frac{1}{x} \right) &= \left(\lim_{x \rightarrow a} x \right) \left(\lim_{x \rightarrow a} \sin \frac{1}{x} \right) \\ &= a \sin \frac{1}{a} = f(a). \end{aligned}$$

From Example 8 in Section 2.3,

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 = f(0).$$

Hence by Definition (2.23), f is continuous at every real number.

The graph of $y = f(x)$ is symmetric with respect to the y -axis, since substitution of $-x$ for x gives us

$$y = -x \sin \frac{1}{-x} = -x \left(-\sin \frac{1}{x} \right) = x \sin \frac{1}{x}.$$

Thus we may restrict our discussion to points (x, y) with $x \geq 0$, since the remainder of the graph can be determined by a reflection through the y -axis.

To find the x -intercepts we solve the equation $x \sin (1/x) = 0$, obtaining $x = 0$ and $1/x = n\pi$, where n is any integer. In particular, in the interval $[0, 1]$ the graph has an infinite number of x -intercepts of the form $1/n\pi$. Several of these are

$$\frac{1}{\pi}, \quad \frac{1}{2\pi}, \quad \frac{1}{3\pi}, \quad \frac{1}{4\pi}, \quad \text{and} \quad \frac{1}{5\pi}.$$

It is of interest to note that the function f is continuous on the interval $[0, 1]$, but its graph cannot be sketched without lifting the pencil from the paper since there is no “first” x -intercept greater than 0.

It is not difficult to show that

$$f(x) < 0 \quad \text{if} \quad \frac{1}{2\pi} < x < \frac{1}{\pi}$$

and
$$f(x) > 0 \quad \text{if} \quad \frac{1}{3\pi} < x < \frac{1}{2\pi}.$$

Similar inequalities can be obtained if $1/((n+1)\pi) < x < 1/n\pi$, where n is any positive integer.

If $x \neq 0$, then $|\sin (1/x)| \leq 1$ and hence

$$|f(x)| = \left| x \sin \frac{1}{x} \right| = |x| \left| \sin \frac{1}{x} \right| \leq |x|.$$

It follows that the graph of f lies between (and on) the lines $y = x$ and $y = -x$. The graph intersects these lines if $|f(x)| = |x|$, that is, if $|\sin(1/x)| = 1$. Observe that $|\sin (1/x)| = 1$ if

$$\frac{1}{x} = \frac{\pi}{2} + n\pi = \frac{\pi(1+2n)}{2}, \quad \text{or} \quad x = \frac{2}{\pi(1+2n)}$$

where n is any integer. The x -coordinates of several such points of intersection are

$$\frac{2}{\pi}, \quad \frac{2}{3\pi}, \quad \frac{2}{5\pi}, \quad \frac{2}{7\pi}, \quad \text{and} \quad \frac{2}{9\pi}.$$

In Exercise 58 the reader is asked to show that the graph of f is tangent to $y = x$ or $y = -x$ at each of these points, and that the graph has a horizontal tangent line at $(x, f(x))$ if $\tan (1/x) = 1/x$.

Using the information we have obtained leads to the graph in Figure 8.2.

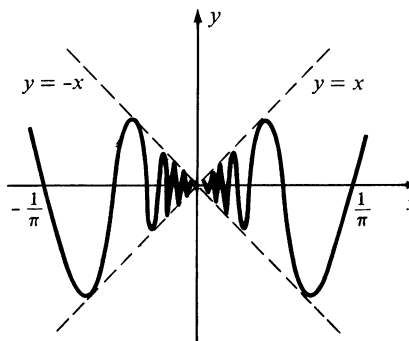


FIGURE 8.2

Example 4 A revolving beacon in a lighthouse that is 200 ft from the nearest point P on a straight shoreline makes one revolution every 15 seconds. Find the rate at which a ray from the light moves along the shore at a point 400 ft from P .

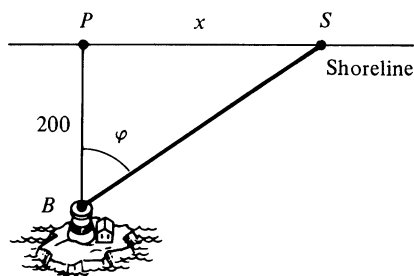


FIGURE 8.3

Solution The problem is diagrammed in Figure 8.3, where B denotes the position of the beacon and φ is the angle between BP and a ray to a point S on the shore x units from P .

Since the light revolves four times per minute, the angle φ changes at a rate of 8π radians per minute, that is, $d\varphi/dt = 8\pi$. Using triangle PBS we see that $\tan \varphi = x/200$, or $x = 200 \tan \varphi$. Consequently,

$$\frac{dx}{dt} = 200 \sec^2 \varphi \frac{d\varphi}{dt} = (200 \sec^2 \varphi)(8\pi) = 1600\pi \sec^2 \varphi.$$

If $x = 400$, then

$$\tan \varphi = 400/200 = 2, \quad \text{and} \quad \sec \varphi = \sqrt{1 + \tan^2 \varphi} = \sqrt{1 + 4} = \sqrt{5}.$$

Hence,
$$\frac{dx}{dt} = 1600\pi(\sqrt{5})^2 = 8000\pi \text{ ft/min.}$$

8.1 Exercises

- 1 Verify the formulas for $D_x \tan u$ and $D_x \csc u$.
- 2 Verify the graphs of the cosecant, secant, tangent, and cotangent functions. In each case discuss continuity, determine where the function increases or decreases, and find the local extrema. Discuss concavity, find the points of inflection, and locate the vertical asymptotes.

In Exercises 3–34 find $f'(x)$ if $f(x)$ equals the given expression.

3 $\tan(8x + 3)$

4 $\cot(x/2)$

5 $\sec \sqrt{x-1}$

6 $\csc(x^2 + 4)$

7 $\cot(x^3 - 2x)$

8 $\tan \sqrt[3]{5-6x}$

- 9 $\cos 3x^2 + \cos^2 3x$ 10 $\tan^3 6x$
 11 $\csc^2 2x$ 12 $\sec e^{-2x}$
 13 $x^2 \csc 5x$ 14 $x \csc(1/x)$
 15 $\tan^2 x \sec^3 x$ 16 $x^2 \sec^3 4x$
 17 $(\sin 5x - \cos 5x)^5$ 18 $\sin \sqrt{x} + \sqrt{\sin x}$
 19 $\cot^3(3x + 1)$ 20 $e^{\cos 2x}$
 21 $(\cos 4x)/(1 - \sin 4x)$ 22 $(\sec 2x)/(\tan 2x + 1)$
 23 $\ln |\csc x + \cot x|$ 24 $\sin(2x + 3)^4$
 25 $e^{-3x} \tan \sqrt{x}$ 26 $\ln \csc^2 3x$
 27 $\ln \ln \sec 2x$ 28 $\csc(\cot 4x)$
 29 $\tan^3 2x - \sec^3 2x$ 30 $(\tan 2x - \sec 2x)^3$
 31 $\frac{\csc 3x}{x^3 + 1}$ 32 $4x^3 - x^2 \cot^3(1/x)$
 33 $x^{\cot x}$ 34 $(\tan x)^{3x}$

In Exercises 35–40 find dy/dx and d^2y/dx^2 .

- 35 $y = \sec^2 3x$ 36 $y = \cot^3 5x$
 37 $y = \sin x - x \cos x$ 38 $y = \ln \tan x$
 39 $y = \sqrt{\tan x}$ 40 $y = \frac{\cos x - 1}{\cos x + 1}$

In Exercises 41–44 use implicit differentiation to find y' .

- 41 $y = x \csc y$ 42 $xy = \tan xy$
 43 $e^x \cot y = xe^{2y}$ 44 $y^2 + 1 = x^2 \sec y$

In Exercises 45 and 46 find the local extrema of f for $0 \leq x \leq 2\pi$.

- 45 $f(x) = e^{-x} \sec x$ 46 $f(x) = e^{-2x} \tan x$

In Exercises 47 and 48 find the absolute maximum and minimum values of f on the given interval.

- 47 $f(x) = \tan x - \sqrt{2} \sec x, [0, \pi/3]$
 48 $f(x) = 3 \cot x + 4x, [\pi/6, 5\pi/6]$

In Exercises 49 and 50 find equations of the tangent and normal lines to the graph of the given equation at the point P .

- 49 $y = (1/8) \csc^3 x, P(\pi/6, 1)$
 50 $y = \tan x - \sqrt{2} \sin x; P(3\pi/4, -2)$

- 51 At a point 20 ft from the base of a flagpole, the angle of elevation of the top of the pole is measured as 60° , with a possible error of $15'$. Use differentials to approximate the error in the calculated height of the pole.
 52 Use differentials to approximate the change in $\cot x$ if x changes from 45° to 46° .
 53 An airplane at an altitude of 20,000 ft is flying at a constant speed on a line that will take it directly over an observer on the ground. If, at a given instant, the observer notes that the angle of elevation of the airplane is 45° and is increasing at a rate of 1° per second, find the speed of the airplane.
 54 A missile is fired vertically from a point which is 5 mi from a tracking station and at the same elevation (see figure). For the first 20 seconds of flight its angle of elevation θ changes at a constant rate of 2° per second. Find the velocity of the missile when the angle of elevation is 30° .

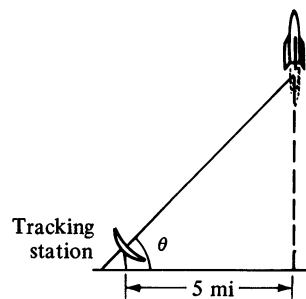


FIGURE FOR EXERCISE 54

- 55 A billboard 20 ft high is located on top of a building, with its lower edge 60 ft above the level of a viewer's eye (see figure). How far from a point directly below the sign should a viewer stand to maximize the angle θ between the lines of sight of the top and bottom of the billboard? (This angle should result in the best view of the billboard).

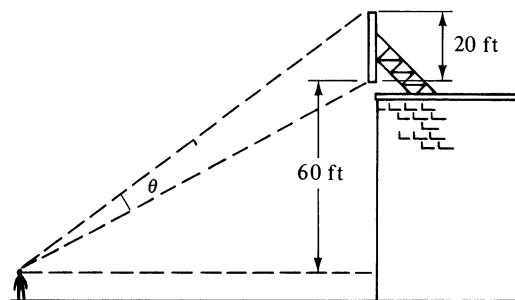


FIGURE FOR EXERCISE 55

- 56 A man on a small island I , which is k mi from the closest point A on a straight shoreline, wishes to reach a camp that is d mi downshore from A (see figure). He plans to swim to some point P on shore and then walk the rest of the way. Suppose that c_1 calories per mile are expended while swimming and c_2 calories per mile while walking, where $c_1 > c_2$.

- (a) Find a formula for the total number c of calories expended in completing the trip.
 (b) Show that the minimum value of c depends only on angle AIP , and not on the distance k or d .

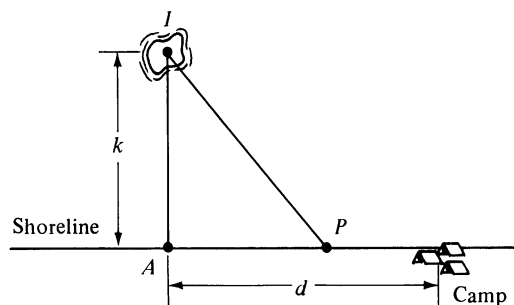


FIGURE FOR EXERCISE 56

- 57 A stuntman falls from a hot-air balloon that is hovering at a constant altitude, 100 ft above a lake. A television camera on shore, 200 ft from a point directly below the

balloon, follows his descent (see figure). At what rate is the angle of elevation θ of the camera changing 2 seconds after the stuntman jumps? (Neglect the height of the camera).

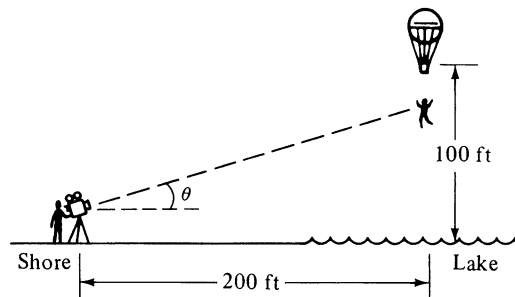


FIGURE FOR EXERCISE 57

- 58 Let f be the function defined in Example 3.
- (a) Prove that the graph of f has a horizontal tangent line at $(x, f(x))$ if $\tan(1/x) = 1/x$. Show that there are an infinite number of such points by sketching the graphs of $y = \tan t$ and $y = t$ on the same coordinate axes.
- (b) Show that the lines $y = x$ and $y = -x$ are tangent to the graph of f at an infinite number of points.

8.2 Integrals of Trigonometric Functions

Formulas for indefinite integrals of important trigonometric expressions are listed in the next theorem, where $u = g(x)$, and g is differentiable.

Theorem (8.2)

$$\int \sin u \, du = -\cos u + C$$

$$\int \cos u \, du = \sin u + C$$

$$\int \tan u \, du = \ln |\sec u| + C$$

$$\int \cot u \, du = \ln |\sin u| + C$$

$$\int \sec u \, du = \ln |\sec u + \tan u| + C$$

$$\int \csc u \, du = \ln |\csc u - \cot u| + C$$

$$\int \sec^2 u \, du = \tan u + C$$

$$\int \csc^2 u \, du = -\cot u + C$$

$$\int \sec u \tan u \, du = \sec u + C$$

$$\int \csc u \cot u \, du = -\csc u + C$$

Proof We shall verify several of the formulas and leave the remainder as exercises. It suffices to consider the case $u = x$, since the more general formulas then follow from Theorem (5.31).

An integration formula for $\tan x$ may be obtained by first writing

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx.$$

If we now let $v = \cos x$, then $dv = -\sin x \, dx$ and hence

$$\begin{aligned} \int \tan x \, dx &= - \int \frac{1}{\cos x} (-\sin x) \, dx = - \int \frac{1}{v} \, dv \\ &= -\ln |v| + C. \end{aligned}$$

Consequently, if $\cos x \neq 0$, then

$$\int \tan x \, dx = -\ln |\cos x| + C.$$

Since $-\ln |\cos x| = \ln (1/|\cos x|) = \ln |\sec x|$, we may also write

$$\int \tan x \, dx = \ln |\sec x| + C.$$

Similarly, since $\cot x = \cos x/\sin x$,

$$\int \cot x \, dx = \int \frac{1}{\sin x} \cos x \, dx.$$

Letting $v = \sin x$ and $dv = \cos x \, dx$ gives us

$$\int \cot x \, dx = \int \frac{1}{v} \, dv = \ln |v| + C = \ln |\sin x| + C.$$

An integral formula for $\sec x$ may be found by using the following technique:

$$\begin{aligned} \int \sec x \, dx &= \int \sec x \frac{\sec x + \tan x}{\sec x + \tan x} \, dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx. \end{aligned}$$

If $v = \tan x + \sec x$, then $dv = (\sec^2 x + \sec x \tan x) \, dx$ and formal substitution gives us

$$\begin{aligned} \int \sec x \, dx &= \int \frac{1}{v} \, dv \\ &= \ln |v| + C \\ &= \ln |\sec x + \tan x| + C. \end{aligned}$$

To verify a formula such as $\int \sec^2 x \, dx = \tan x + C$, it is sufficient to note that $D_x \tan x = \sec^2 x$. $\square$

Example 1 Find $\int x \cot x^2 dx$.

Solution If $u = x^2$, then $du = 2x dx$ and the form $\int \cot u du$ may be obtained by introducing the factor 2 in the integrand. This leads to

$$\begin{aligned}\int x \cot x^2 dx &= \frac{1}{2} \int \cot x^2 (2x) dx \\ &= \frac{1}{2} \int \cot u du = \frac{1}{2} \ln |\sin u| + C \\ &= \frac{1}{2} \ln |\sin x^2| + C. \quad \blacksquare\end{aligned}$$

Example 2 Find $\int \sec x (\sec x + \tan x) dx$.

Solution We may proceed as follows:

$$\begin{aligned}\int \sec x (\sec x + \tan x) dx &= \int (\sec^2 x + \sec x \tan x) dx \\ &= \int \sec^2 x dx + \int \sec x \tan x dx \\ &= \tan x + \sec x + C. \quad \blacksquare\end{aligned}$$

Example 3 Find $\int \left(\frac{1}{\sqrt{x}}\right) \csc^2 \sqrt{x} dx$.

Solution If

$$u = \sqrt{x}, \quad \text{then} \quad du = \left(\frac{1}{2\sqrt{x}}\right) dx$$

and the integrand can be written in the form $\int \csc^2 u du$ by introducing the factor $\frac{1}{2}$. Thus,

$$\begin{aligned}\int \left(\frac{1}{\sqrt{x}}\right) \csc^2 \sqrt{x} dx &= 2 \int \csc^2 \sqrt{x} \left(\frac{1}{2\sqrt{x}}\right) dx \\ &= 2 \int \csc^2 u du = -2 \cot u + C \\ &= -2 \cot \sqrt{x} + C. \quad \blacksquare\end{aligned}$$

Example 4 Evaluate $\int_0^{\pi/2} \tan \frac{x}{2} dx$.

Solution If $u = x/2$, then $du = \frac{1}{2} dx$. Making this change of variable and noting that $u = 0$ if $x = 0$, and $u = \pi/4$ if $x = \pi/2$, we obtain

$$\begin{aligned}\int_0^{\pi/2} \tan \frac{x}{2} dx &= 2 \int_0^{\pi/2} \tan \frac{x}{2} \cdot \frac{1}{2} dx \\ &= 2 \int_0^{\pi/4} \tan u du = 2 \ln \sec u \Big|_0^{\pi/4}.\end{aligned}$$

It is unnecessary to employ the absolute value sign as in Theorem (8.2) since $\sec u$ is positive if u is between 0 and $\pi/4$. Since $\ln \sec \pi/4 = \ln \sqrt{2} = (\frac{1}{2}) \ln 2$, and $\ln \sec 0 = \ln 1 = 0$, it follows that

$$\int_0^{\pi/2} \tan \frac{x}{2} dx = 2 \cdot \frac{1}{2} \ln 2 = \ln 2. \quad \blacksquare$$

Example 5 Evaluate $\int e^{2x} \sec e^{2x} dx$.

Solution If we let $u = e^{2x}$, then $du = 2e^{2x} dx$, and

$$\begin{aligned} \int e^{2x} \sec e^{2x} dx &= \frac{1}{2} \int \sec e^{2x} (2e^{2x}) dx \\ &= \frac{1}{2} \int \sec u du \\ &= \frac{1}{2} \ln |\sec u + \tan u| + C \\ &= \frac{1}{2} \ln |\sec e^{2x} + \tan e^{2x}| + C. \quad \blacksquare \end{aligned}$$

Example 6 Evaluate $\int (\csc x - 1)^2 dx$.

$$\begin{aligned} \text{Solution} \quad \int (\csc x - 1)^2 dx &= \int (\csc^2 x - 2 \csc x + 1) dx \\ &= \int \csc^2 x dx - 2 \int \csc x dx + \int dx \\ &= -\cot x - 2 \ln |\csc x - \cot x| + x + C. \quad \blacksquare \end{aligned}$$

Additional methods for integrating trigonometric expressions will be discussed in Chapter 9.

8.2 Exercises

Evaluate the integrals in Exercises 1–28.

1 $\int \csc 4x dx$

2 $\int \sec^2 5x dx$

7 $\int \frac{1}{\cos 2x} dx$

8 $\int \frac{\cot \sqrt[3]{x}}{\sqrt[3]{x^2}} dx$

3 $\int \tan 3x \sec 3x dx$

4 $\int x^2 \cot x^3 \csc x^3 dx$

9 $\int x \csc^2 (x^2 + 1) dx$

10 $\int (x + \csc 8x) dx$

5 $\int (\tan 3x + \sec 3x) dx$

6 $\int \frac{1}{\sec 2x} dx$

11 $\int \cot 6x \sin 6x dx$

12 $\int \sin 2x \tan 2x dx$

$$13 \int_0^{\pi/4} \tan x \sec^2 x \, dx \qquad 14 \int \csc^2 x \cot x \, dx$$

$$15 \int \frac{\tan^2 2x}{\sec 2x} \, dx \qquad 16 \int_{\pi/6}^{\pi/2} \frac{\cos^2 x}{\sin x} \, dx$$

$$17 \int_{\pi/6}^{\pi} \sin x \cos x \, dx \qquad 18 \int \frac{\cos^2 x}{\csc x} \, dx$$

$$19 \int \frac{1 - \sin x}{x + \cos x} \, dx \qquad 20 \int \frac{e^x}{\cos e^x} \, dx$$

$$21 \int_{\pi/4}^{\pi/3} \frac{1 + \sin x}{\cos^2 x} \, dx \qquad 22 \int_0^{\pi/4} (1 + \sec x)^2 \, dx$$

$$23 \int e^x(1 + \tan e^x) \, dx \qquad 24 \int (\csc^2 x)2^{\cot x} \, dx$$

$$25 \int \frac{e^{\cos x}}{\csc x} \, dx \qquad 26 \int \frac{\tan e^{-3x}}{e^{3x}} \, dx$$

$$27 \int \frac{\sec^2 x}{2 \tan x + 1} \, dx \qquad 28 \int \frac{\sec x \tan x}{1 + 3 \sec x} \, dx$$

29 Find the area of the region under the graph of $y = \cot(x/2)$ from $x = \pi/3$ to $x = \pi/2$.

30 Find the area of the region under the graph of $y = 2 \tan x$ from $x = 0$ to $x = \pi/4$.

31 Find the area of the region bounded by the graphs of $y = \sec x$, $y = x$, $x = -\pi/4$, and $x = \pi/4$.

32 Find the area of the region bounded by the graphs of $y = \sin x$, $y = \csc x$, $x = \pi/4$, and $x = \pi/2$.

33 The region bounded by the graphs of the equations $y = \sec x$, $x = -\pi/3$, $x = \pi/3$, and $y = 0$ is revolved about the x -axis. Find the volume of the resulting solid.

34 The region between the graphs of $y = \tan(x^2)$ and the x -axis, from $x = 0$ to $x = \sqrt{\pi/2}$, is revolved about the y -axis. Find the volume of the resulting solid.

35 Verify the formula for $\int \csc u \, du$.

36 Verify the formula for $\int \cot u \, du$.

Verify the formulas in Exercises 37 and 38 by evaluating the integrals in two different ways. How can the answers be reconciled?

$$37 \int \tan x \sec^2 x \, dx = \frac{1}{2} \tan^2 x + C = \frac{1}{2} \sec^2 x + D.$$

$$38 \int \cot x \csc^2 x \, dx = -\frac{1}{2} \cot^2 x + C = -\frac{1}{2} \csc^2 x + D.$$

In Exercises 39 and 40, (a) set up an integral for finding the arc length L , from A to B , of the graph of the given equation; (b) write out the formula for approximating L by means of Simpson's Rule (5.35) with $n = 4$; (c) use a calculator to find the approximation given by the formula in part (b).

$$39 \quad y = \sec(x/2); \quad A(0, 1), \quad B(\pi/2, \sqrt{2})$$

$$40 \quad y = \tan x; \quad A(0, 0), \quad B(\pi/4, 1)$$

41 The ends of a water trough 6 ft long have the shape of the graph of $y = \sec x$ from $x = -\pi/3$ to $x = \pi/3$, where x is in feet. Suppose the trough is full of water.

(a) Set up a definite integral with respect to y that equals the work required to pump the water out over the top.

(b) Change the integral in (a) to an integral with respect to x and use Simpson's Rule with $n = 4$ on the interval $[-\pi/3, \pi/3]$ to approximate the work required.

42 Given the full water trough described in Exercise 41, (a) set up a definite integral with respect to y that equals the total force on one of the ends and (b) change the integral in (a) to an integral with respect to x and use the Trapezoidal Rule with $n = 8$ on the interval $[-\pi/3, \pi/3]$ to approximate the total force.

8.3 The Inverse Trigonometric Functions

As in Section 7.1, if f is a one-to-one function with domain Y and range X , then its inverse function f^{-1} has domain X and range Y . Moreover,

$$y = f^{-1}(x) \quad \text{if and only if} \quad f(y) = x$$

for every x in X and every y in Y . Since the trigonometric functions are not one-to-one, they do not have inverses. However, by restricting the domains

it is possible to obtain functions that vary in the same way as the trigonometric functions (over the smaller domains), but do possess inverse functions.

Let us first consider the sine function whose domain is $\mathbb{R}$ and range is the set of real numbers in the interval $[-1, 1]$. The sine function is not one-to-one since, for example, numbers such as $\pi/6$, $5\pi/6$, and $-7\pi/6$ lead to the same functional value, $\frac{1}{2}$. It is easy to find a subset S of $\mathbb{R}$ with the property that as x ranges through S , $\sin x$ takes on each value between -1 and 1 once and only once. It is convenient to choose the interval $[-\pi/2, \pi/2]$ for S . The new function obtained by restricting the domain of the sine function to $[-\pi/2, \pi/2]$ is continuous and increasing and hence, by Theorem (7.26), has an inverse function which is continuous and increasing on $[-1, 1]$. This leads to the following definition.

Definition (8.3)

The **inverse sine function**, denoted $\sin^{-1}$, is defined by

$$y = \sin^{-1} x \quad \text{if and only if} \quad \sin y = x$$

where $-1 \leq x \leq 1$ and $-\pi/2 \leq y \leq \pi/2$.

It is also customary to refer to this function as the **arcsine function** and use $\arcsin x$ in place of $\sin^{-1} x$. We shall employ both notations in our work. The -1 in $\sin^{-1}$ is not to be regarded as an exponent, but rather as a means of denoting this inverse function. Observe that by Definition (8.3),

$$-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2} \quad \text{or} \quad -\frac{\pi}{2} \leq \arcsin x \leq \frac{\pi}{2}.$$

It follows from (8.3) that the graph of $y = \sin^{-1} x$ (or $y = \arcsin x$) may be found by sketching the graph of $x = \sin y$, where $-\pi/2 \leq y \leq \pi/2$ (see Figure 8.4).

Since $\sin$ and $\sin^{-1}$ are inverse functions of one another,

$$\sin^{-1}(\sin x) = x \quad \text{if} \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$\text{and} \quad \sin(\sin^{-1} x) = x \quad \text{if} \quad -1 \leq x \leq 1.$$

These formulas may also be written

$$\arcsin(\sin x) = x \quad \text{and} \quad \sin(\arcsin x) = x.$$

Example 1 Find $\sin^{-1}(\sqrt{2}/2)$ and $\arcsin(-\frac{1}{2})$.

Solution If $y = \sin^{-1}(\sqrt{2}/2)$, then $\sin y = \sqrt{2}/2$ and consequently $y = \pi/4$. Note that it is essential to choose y in the interval $[-\pi/2, \pi/2]$. A number such as $3\pi/4$ is incorrect, even though $\sin(3\pi/4) = \sqrt{2}/2$.

In like manner, if $y = \arcsin(-\frac{1}{2})$, then $\sin y = -\frac{1}{2}$ and hence $y = -\pi/6$. ■

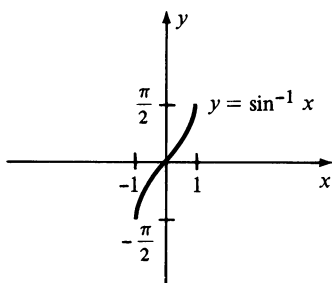


FIGURE 8.4

Similar discussions may be given for the other trigonometric functions. If the domain of the cosine function is restricted to the interval $[0, \pi]$ we obtain a continuous, decreasing function that has a continuous decreasing inverse function by Theorem (7.27). This leads to the next definition.

Definition (8.4)

The **inverse cosine function**, denoted by $\cos^{-1}$, is defined by

$$y = \cos^{-1} x \quad \text{if and only if} \quad \cos y = x$$

where $-1 \leq x \leq 1$ and $0 \leq y \leq \pi$.

The inverse cosine function is also referred to as the **arccosine function** and the notation $\arccos x$ is used interchangeably with $\cos^{-1} x$. Note that

$$\cos^{-1}(\cos x) = \arccos(\cos x) = x \quad \text{if} \quad 0 \leq x \leq \pi$$

$$\cos(\cos^{-1} x) = \cos(\arccos x) = x \quad \text{if} \quad -1 \leq x \leq 1.$$

If we restrict the domain of the tangent function to the open interval $(-\pi/2, \pi/2)$, then a one-to-one function is obtained. We may, therefore, adopt the following definition.

Definition (8.5)

The **inverse tangent function** or **arctangent function**, denoted by $\tan^{-1}$ or $\arctan$, is defined by

$$y = \tan^{-1} x = \arctan x \quad \text{if and only if} \quad \tan y = x$$

where x is any real number and $-\pi/2 < y < \pi/2$.

The domain of the arctangent function is $\mathbb{R}$ and the range is the open interval $(-\pi/2, \pi/2)$.

The graphs of the inverse cosine and inverse tangent functions are sketched in Figure 8.5.

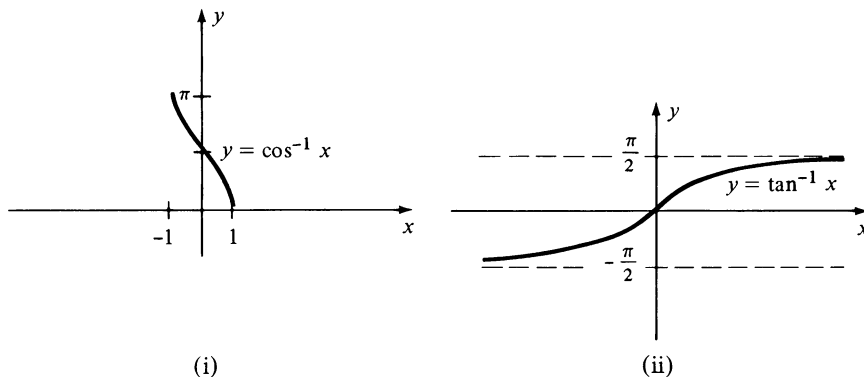


FIGURE 8.5

Example 2 Find $\sec(\arctan \frac{2}{3})$ without using tables or calculators.

Solution If $y = \arctan \frac{2}{3}$, then $\tan y = \frac{2}{3}$. Since $\sec^2 y = 1 + \tan^2 y$ and $0 < y < \pi/2$, we may write

$$\begin{aligned}\sec y &= \sqrt{1 + \tan^2 y} = \sqrt{1 + \left(\frac{2}{3}\right)^2} \\ &= \sqrt{1 + \frac{4}{9}} = \sqrt{\frac{13}{9}} = \frac{\sqrt{13}}{3}.\end{aligned}$$

Hence, $\sec(\arctan \frac{2}{3}) = \sec y = \sqrt{13}/3$. ■

If y varies through the two intervals $[0, \pi/2)$ and $[\pi, 3\pi/2)$, then $\sec y$ takes on each of its values once and only once and an inverse function may be defined.

Definition (8.6)

The **inverse secant** or **arcsecant function**, denoted by $\sec^{-1}$ or arcsec , is defined by

$$y = \sec^{-1} x = \operatorname{arcsec} x \quad \text{if and only if} \quad \sec y = x$$

where $|x| \geq 1$ and y is in $[0, \pi/2)$ or in $[\pi, 3\pi/2)$.

The graph of $y = \sec^{-1} x$ is sketched in Figure 8.6. The main reason for choosing y as in Definition (8.6) instead of the more natural intervals $[0, \pi/2)$ and $(\pi/2, \pi]$ is that the differentiation formula for the inverse secant is simpler.

The following examples illustrate some of the manipulations that may be carried out with inverse trigonometric functions.

Example 3 If $-1 \leq x \leq 1$, rewrite $\cos(\sin^{-1} x)$ as an algebraic expression in x .

Solution Let $y = \sin^{-1} x$, so that $\sin y = x$. We wish to express $\cos y$ in terms of x . Since $-\pi/2 \leq y \leq \pi/2$, it follows that $\cos y \geq 0$, and hence

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}.$$

Consequently,

$$\cos(\sin^{-1} x) = \sqrt{1 - x^2}.$$

Example 4 Verify the identity

$$\frac{1}{2} \cos^{-1} x = \tan^{-1} \sqrt{\frac{1-x}{1+x}}$$

where $|x| < 1$.

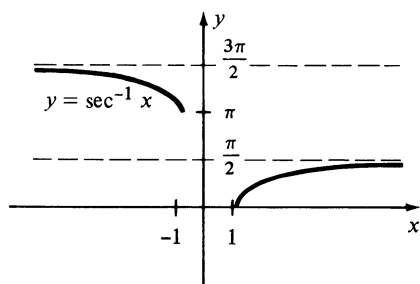


FIGURE 8.6

Solution Let $y = \cos^{-1} x$. We wish to show that

$$\frac{y}{2} = \tan^{-1} \sqrt{\frac{1-x}{1+x}}.$$

By a half-angle formula,

$$\left| \tan \frac{y}{2} \right| = \sqrt{\frac{1 - \cos y}{1 + \cos y}}.$$

Since $y = \cos^{-1} x$ and $|x| < 1$, it follows that $0 < y < \pi$ and hence that $0 < (y/2) < \pi/2$. Consequently, $\tan (y/2) > 0$, and we may drop the absolute value sign, obtaining

$$\tan \frac{y}{2} = \sqrt{\frac{1 - \cos y}{1 + \cos y}}.$$

Using the fact that $\cos y = x$ gives us

$$\tan \frac{y}{2} = \sqrt{\frac{1-x}{1+x}}$$

$$\text{or, equivalently,} \quad \frac{y}{2} = \tan^{-1} \sqrt{\frac{1-x}{1+x}}$$

which is what we wished to show. ■

8.3 Exercises

Find the exact values in Exercises 1–18 without the use of tables or calculators.

1 (a) $\sin^{-1} \sqrt{3}/2$
(b) $\sin^{-1} (-\sqrt{3}/2)$

2 (a) $\sin^{-1} 0$
(b) $\arccos 0$

3 (a) $\cos^{-1} \sqrt{2}/2$
(b) $\cos^{-1} (-\sqrt{2}/2)$

4 (a) $\arcsin (-1)$
(b) $\cos^{-1} (-1)$

5 (a) $\tan^{-1} \sqrt{3}$
(b) $\arctan (-\sqrt{3})$

6 (a) $\tan^{-1} (-1)$
(b) $\arccos (1/2)$

7 $\sin [\cos^{-1} \sqrt{3}/2]$

8 $\cos [\sin^{-1} 0]$

9 $\sin (\arccos \frac{3}{5})$

10 $\tan [\tan^{-1} 10]$

11 $\arcsin (\sin \sqrt{5})$

12 $\tan^{-1} (\cos 0)$

13 $\cos [\sin^{-1} \frac{3}{5} + \tan^{-1} \frac{4}{3}]$

14 $\sin [\arcsin \frac{1}{2} + \arccos 0]$

15 $\tan [\arctan \frac{3}{4} + \arccos \frac{3}{5}]$

16 $\cos [2 \sin^{-1} \frac{8}{17}]$

17 $\sin [2 \arccos (-\frac{4}{5})]$

18 $\sin (\arctan \frac{1}{2} - \arccos \frac{4}{5})$

Rewrite the expressions in Exercises 19–22 as algebraic expressions in x .

19 $\sin (\tan^{-1} x)$

20 $\tan (\arccos x)$

21 $\cos (\frac{1}{2} \arccos x)$

22 $\cos (2 \tan^{-1} x)$

Verify the identities in Exercises 23–30.

23 $\sin^{-1} x + \cos^{-1} x = \pi/2$ (Hint: Let $\alpha = \sin^{-1} x$, $\beta = \cos^{-1} x$, and consider $\sin(\alpha + \beta)$.)

24 $\arctan x + \arctan(1/x) = \pi/2$, $x > 0$

25 $\arcsin \frac{2x}{1+x^2} = 2 \arctan x$, $|x| \leq 1$

26 $2 \cos^{-1} x = \cos^{-1}(2x^2 - 1)$, $0 \leq x \leq 1$

27 $\sin^{-1}(-x) = -\sin^{-1} x$

28 $\arccos(-x) = \pi - \arccos x$

29 $\sin^{-1} x = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$

30 $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$, where $|x| < 1$ and $|y| < 1$.

31 Define $\cot^{-1}$ by restricting the domain of $\cot$ to the interval $(0, \pi)$.

32 Define $\csc^{-1}$ by restricting the domain of $\csc$ to $(0, \pi/2]$ and $(\pi, 3\pi/2]$.

Sketch the graphs of the equations in Exercises 33–42.

33 $y = \sin^{-1} 2x$

34 $y = \cos^{-1}(x/2)$

35 $y = \frac{1}{2} \sin^{-1} x$

36 $y = 2 \cos^{-1} x$

37 $y = 2 \tan^{-1} x$

38 $y = \tan^{-1} 2x$

39 $y = \sin(\sin^{-1} x)$

40 $y = \sin(\arccos x)$

41 $y = \sin^{-1}(\sin x)$

42 $y = \arccos(\cos x)$

Show that the equations in Exercises 43 and 44 are *not* identities.

43 $\tan^{-1} x = 1/\tan x$

44 $(\arcsin x)^2 + (\arccos x)^2 = 1$.

8.4 Derivatives and Integrals Involving Inverse Trigonometric Functions

In this section we shall concentrate on the inverse sine, cosine, tangent, and secant functions. Formulas for their derivatives and for integrals that result in inverse trigonometric functions are listed in the next three theorems, where $u = g(x)$ and x is restricted to values for which the indicated expressions are defined.

Theorem (8.7)

$$\begin{aligned} D_x \sin^{-1} u &= \frac{1}{\sqrt{1-u^2}} D_x u & D_x \cos^{-1} u &= -\frac{1}{\sqrt{1-u^2}} D_x u \\ D_x \tan^{-1} u &= \frac{1}{1+u^2} D_x u & D_x \sec^{-1} u &= \frac{1}{u\sqrt{u^2-1}} D_x u \end{aligned}$$

Proof As usual, we shall consider the special case $u = x$, since the general formula may then be obtained by applying the Chain Rule.

If we let $f(x) = \sin x$ and $g(x) = \sin^{-1} x$ in Theorem (7.28), then it follows that the inverse sine function g is differentiable if $|x| < 1$. We shall use implicit differentiation to find $g'(x)$. First note that by Definition (8.3),

$$y = \sin^{-1} x \quad \text{if and only if} \quad \sin y = x$$

where $-1 < x < 1$, and $-\pi/2 < y < \pi/2$. Differentiating the last equation implicitly, we obtain

$$\cos y \, D_x y = 1$$

and hence
$$D_x y = D_x \sin^{-1} x = \frac{1}{\cos y}.$$

Since $-\pi/2 < y < \pi/2$, $\cos y$ is positive and, therefore,

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}.$$

Substitution in the previous formula for $D_x y$ gives us

$$D_x \sin^{-1} x = \frac{1}{\sqrt{1 - x^2}}$$

provided $|x| < 1$. The inverse sine function is not differentiable at ± 1 . This fact is geometrically evident from Figure 8.4, since vertical tangent lines occur at the endpoints of the graph.

In similar fashion, to find the derivative of the inverse cosine function we begin with the equivalent equations

$$y = \cos^{-1} x \quad \text{and} \quad \cos y = x$$

where $|x| < 1$ and $0 < y < \pi$. Differentiating the second equation implicitly gives us

$$-\sin y \, D_x y = 1$$

and, therefore,
$$D_x y = D_x \cos^{-1} x = -\frac{1}{\sin y}.$$

Since $0 < y < \pi$, $\sin y = \sqrt{1 - \cos^2 y} = \sqrt{1 - x^2}$. Consequently, if $|x| < 1$, then

$$D_x \cos^{-1} x = -\frac{1}{\sqrt{1 - x^2}}.$$

It follows from Theorem (7.28) that the inverse tangent function is differentiable at every real number. If we consider the equivalent equations

$$y = \tan^{-1} x \quad \text{and} \quad \tan y = x$$

where $-\pi/2 < y < \pi/2$, and differentiate the second equation implicitly, the result is

$$\sec^2 y \, D_x y = 1.$$

Consequently,
$$D_x y = D_x \tan^{-1} x = \frac{1}{\sec^2 y}.$$

Using the fact that $\sec^2 y = 1 + \tan^2 y = 1 + x^2$ gives us

$$D_x \tan^{-1} x = \frac{1}{1 + x^2}.$$

Finally, consider the equivalent equations

$$y = \sec^{-1} x \quad \text{and} \quad \sec y = x$$

where y is either in $[0, \pi/2)$ or $[\pi, 3\pi/2)$. Differentiating the second equation implicitly gives us

$$\sec y \tan y D_x y = 1$$

or, if $\tan y \neq 0$,

$$D_x y = D_x \sec^{-1} x = \frac{1}{\sec y \tan y}.$$

Using the fact that $\tan y = \sqrt{\sec^2 y - 1} = \sqrt{x^2 - 1}$, we have

$$D_x \sec^{-1} x = \frac{1}{x\sqrt{x^2 - 1}}$$

provided $|x| > 1$. The inverse secant function is not differentiable at $x = \pm 1$. Indeed, the graph has vertical tangent lines at the points with these x -coordinates (see Figure 8.6). $\square$

Example 1 Find dy/dx if $y = \sin^{-1} 3x - \cos^{-1} 3x$.

Solution Applying Theorem (8.7) with $u = 3x$,

$$\frac{dy}{dx} = \frac{3}{\sqrt{1 - 9x^2}} + \frac{3}{\sqrt{1 - 9x^2}} = \frac{6}{\sqrt{1 - 9x^2}}. \quad \blacksquare$$

Example 2 Find $f'(x)$ if $f(x) = \tan^{-1} e^{2x}$.

Solution Using (8.7) with $u = e^{2x}$,

$$f'(x) = \frac{1}{1 + (e^{2x})^2} D_x e^{2x} = \frac{2e^{2x}}{1 + e^{4x}}. \quad \blacksquare$$

Example 3 Find y' if $y = \sec^{-1}(x^2)$.

Solution Applying Theorem (8.7) with $u = x^2$,

$$y' = \frac{1}{x^2 \sqrt{x^4 - 1}} (2x) = \frac{2}{x \sqrt{x^4 - 1}}. \quad \blacksquare$$

Following the same pattern employed many times before, we may use the differentiation formulas for the inverse trigonometric functions to obtain integration formulas. The proof of the next theorem is left to the reader.

Theorem (8.8)

$$\begin{aligned}\int \frac{1}{\sqrt{1-u^2}} du &= \sin^{-1} u + C \\ \int \frac{1}{1+u^2} du &= \tan^{-1} u + C \\ \int \frac{1}{u\sqrt{u^2-1}} du &= \sec^{-1} u + C\end{aligned}$$

Example 4 Evaluate $\int \frac{e^{2x}}{\sqrt{1-e^{4x}}} dx$.

Solution If we let $u = e^{2x}$, then $du = 2e^{2x} dx$ and the integral may be written as in the first formula of Theorem (8.8). Thus,

$$\begin{aligned}\int \frac{e^{2x}}{\sqrt{1-e^{4x}}} dx &= \frac{1}{2} \int \frac{2e^{2x}}{\sqrt{1-(e^{2x})^2}} dx \\ &= \frac{1}{2} \int \frac{1}{\sqrt{1-u^2}} du \\ &= \frac{1}{2} \sin^{-1} u + C \\ &= \frac{1}{2} \sin^{-1} e^{2x} + C.\end{aligned}$$

The formulas in Theorem (8.8) can be extended as follows, where a is a nonzero real number.

Theorem (8.9)

$$\begin{aligned}\int \frac{1}{\sqrt{a^2-u^2}} du &= \sin^{-1} \frac{u}{a} + C \\ \int \frac{1}{a^2+u^2} du &= \frac{1}{a} \tan^{-1} \frac{u}{a} + C \\ \int \frac{1}{u\sqrt{u^2-a^2}} du &= \frac{1}{a} \sec^{-1} \frac{u}{a} + C\end{aligned}$$

Proof Let us prove the second formula. If $u = g(x)$ and g is differentiable, then by Theorem (8.7),

$$\begin{aligned} D_x \left[\frac{1}{a} \tan^{-1} \frac{g(x)}{a} \right] &= \frac{1}{a} \frac{1}{1 + [g(x)/a]^2} D_x \left[\frac{1}{a} g(x) \right] \\ &= \frac{1}{a} \frac{a^2}{a^2 + [g(x)]^2} \frac{1}{a} g'(x) \\ &= \frac{1}{a^2 + [g(x)]^2} g'(x). \end{aligned}$$

Consequently,

$$\int \frac{1}{a^2 + [g(x)]^2} g'(x) dx = \frac{1}{a} \tan^{-1} \frac{g(x)}{a} + C = \frac{1}{a} \tan^{-1} \frac{u}{a} + C.$$

The verifications of the remaining formulas are left as exercises. $\square$

Example 5 Evaluate $\int \frac{x^2}{5 + x^6} dx$.

Solution If we let $u = x^3$, then $du = 3x^2 dx$ and the given integral may be written as in the second formula of Theorem (8.9). Thus,

$$\begin{aligned} \int \frac{x^2}{5 + x^6} dx &= \frac{1}{3} \int \frac{3x^2}{5 + (x^3)^2} dx \\ &= \frac{1}{3} \int \frac{1}{(\sqrt{5})^2 + u^2} du \\ &= \frac{1}{3} \cdot \frac{1}{\sqrt{5}} \tan^{-1} \frac{u}{\sqrt{5}} + C \\ &= \frac{\sqrt{5}}{15} \tan^{-1} \frac{x^3}{\sqrt{5}} + C. \end{aligned}$$

Example 6 Evaluate $\int \frac{1}{x\sqrt{x^4 - 9}} dx$.

Solution If we let $u = x^2$, then $du = 2x dx$ and the integral may be transformed into the proper form as follows:

$$\begin{aligned} \int \frac{1}{x\sqrt{x^4 - 9}} dx &= \frac{1}{2} \int \frac{1}{x^2 \sqrt{(x^2)^2 - 9}} 2x dx \\ &= \frac{1}{2} \int \frac{1}{u \sqrt{u^2 - 9}} du \\ &= \frac{1}{2} \cdot \frac{1}{3} \sec^{-1} \frac{u}{3} + C \\ &= \frac{1}{6} \sec^{-1} \frac{x^2}{3} + C. \end{aligned}$$

8.4 Exercises

In Exercises 1–24 find $f'(x)$ if $f(x)$ equals the given expression.

- | | |
|---|---|
| 1 $\tan^{-1}(3x - 5)$ | 2 $\sin^{-1}(x/3)$ |
| 3 $\sin^{-1}\sqrt{x}$ | 4 $\tan^{-1}x^2$ |
| 5 $e^{-x} \operatorname{arccsc} e^{-x}$ | 6 $\sqrt{\operatorname{arccsc} 3x}$ |
| 7 $x^2 \arctan x^2$ | 8 $\tan^{-1} \sin 2x$ |
| 9 $(1 + \cos^{-1} 3x)^3$ | 10 $x^2 \sec^{-1} 5x$ |
| 11 $\ln \arctan x^2$ | 12 $\arcsin \ln x$ |
| 13 $1/\sin^{-1} x$ | 14 $\arctan \frac{x+1}{x-1}$ |
| 15 $\sec^{-1} \sqrt{x^2 - 1}$ | 16 $\left(\frac{1}{x} - \arcsin \frac{1}{x}\right)^4$ |
| 17 $(\arctan x)/(x^2 + 1)$ | 18 $\cos^{-1} \cos e^x$ |
| 19 $\sqrt{x} \sec^{-1} \sqrt{x}$ | 20 $e^{2x}/(\sin^{-1} 5x)$ |
| 21 $3 \arcsin x^3$ | 22 $x \arccos \sqrt{4x+1}$ |
| 23 $(\tan x)^{\arctan x}$ | 24 $(\tan^{-1} 4x)e^{\tan^{-1} 4x}$ |

Find y' in Exercises 25 and 26.

- 25 $x^2 + x \sin^{-1} y = ye^x$ 26 $\ln(x + y) = \tan^{-1} xy$

Evaluate the integrals in Exercises 27–42.

- | | |
|--|---|
| 27 $\int_0^4 \frac{1}{x^2 + 16} dx$ | 28 $\int_0^1 \frac{e^x}{1 + e^{2x}} dx$ |
| 29 $\int_0^{\sqrt{2}/2} \frac{x}{\sqrt{1-x^4}} dx$ | 30 $\int_{2/\sqrt{3}}^2 \frac{1}{x\sqrt{x^2-1}} dx$ |
| 31 $\int \frac{\sin x}{\cos^2 x + 1} dx$ | 32 $\int \frac{\cos x}{\sqrt{9 - \sin^2 x}} dx$ |
| 33 $\int \frac{1}{\sqrt{x(1+x)}} dx$ | 34 $\int \frac{1}{e^x \sqrt{1 - e^{-2x}}} dx$ |
| 35 $\int \frac{e^x}{\sqrt{16 - e^{2x}}} dx$ | 36 $\int \frac{\sec x \tan x}{1 + \sec^2 x} dx$ |
| 37 $\int \frac{1}{x\sqrt{x^6 - 4}} dx$ | 38 $\int \frac{x}{\sqrt{36 - x^2}} dx$ |
| 39 $\int \frac{x}{x^2 + 9} dx$ | 40 $\int \frac{1}{x\sqrt{x-1}} dx$ |
| 41 $\int \frac{1}{\sqrt{e^{2x} - 25}} dx$ | 42 $\int \frac{e^x}{\sqrt{4 - e^x}} dx$ |

- 43 Find the area of the region bounded by the graphs of the equations $y = 4/\sqrt{16 - x^2}$, $x = -2$, $x = 2$, and $y = 0$.
- 44 If $f(x) = x^2/(1 + x^6)$, find the area of the region under the graph of f from $x = 0$ to $x = 1$.
- 45 The floor of a storage shed has the shape of a right triangle. The sides opposite and adjacent to an acute angle θ of the triangle are measured as 10 ft and 7 ft, respectively, with a possible error of $\frac{1}{2}$ inch in the 10-foot measurement. Use the differential of an inverse trigonometric function to approximate the error in the calculated value of θ .
- 46 Use differentials to approximate the change in $\arcsin x$ if x changes from 0.25 to 0.26.
- 47 An airplane at an altitude of 5 mi and a speed of 500 mi/hr is flying in a direction away from an observer on the ground. Use inverse trigonometric functions to find the rate at which the angle of elevation is changing when the airplane is over a point 2 mi from the observer.
- 48 A searchlight located $\frac{1}{8}$ mi from the nearest point P on a straight road is trained on an automobile traveling on the road at a rate of 50 mi/hr. Use inverse trigonometric functions to find the rate at which the searchlight is rotating when the car is $\frac{1}{4}$ mi from P .
- 49 A billboard 20 ft high is located on top of a building, with its lower edge 60 ft above the level of a viewer's eye. Use inverse trigonometric functions to find how far from a point directly below the sign a viewer should stand to maximize the angle between the lines of sight of the top and bottom of the billboard (cf. Exercise 55 of Section 8.1).
- 50 Given points $A(3, 1)$ and $B(6, 4)$ on a rectangular coordinate system, find the x -coordinate of the point P on the x -axis such that angle APB has its largest value.
- 51 If $f(x) = \sin^{-1}(\sin x)$ find $f'(x)$, $f''(x)$, and the local extrema of f . Discuss concavity and sketch the graph of f .
- 52 Work Exercise 51 if $f(x) = \arccos(\cos x)$.
- 53 Find equations for the tangent and normal lines to the graph of $y = \sin^{-1}(x - 1)$ at the point $(\frac{3}{2}, \pi/6)$.
- 54 Find the points on the graph of $y = \tan^{-1} 2x$ at which the tangent line is parallel to the line $13y - 2x + 5 = 0$.
- 55 Find the intervals in which the graph of $y = \tan^{-1} x$ is (a) concave upward; (b) concave downward.
- 56 The velocity, at time t , of a point moving on a coordinate line is $(1 + t^2)^{-1}$ ft/sec. If the point is at the origin at $t = 0$, find its position at the instant that the acceleration and velocity have the same absolute value.

- 57 The region bounded by the graphs of the equations $y = e^x$, $y = 1/\sqrt{x^2 + 1}$, and $x = 1$ is revolved about the x -axis. Find the volume of the resulting solid.
- 58 Use differentials to approximate the arc length of the graph of $y = \tan^{-1} x$ from the point $(0, 0)$ to the point $(0.1, \tan^{-1} 0.1)$.
- 59 A missile is fired vertically from a point which is 5 mi from a tracking station and at the same elevation. For the first 20 seconds of flight its angle of elevation changes at a constant rate of 2° per second. Use inverse trigonometric functions to find the velocity of the missile when the angle of elevation is 30° (cf. Exercise 54, Section 8.1).
- 60 Prove that $\int_0^1 \frac{4}{(1+x^2)} dx = \pi$ and then apply Simpson's Rule with $n = 10$ to approximate π .
- 61 Verify the formula for $\int \frac{1}{\sqrt{a^2 - u^2}} du$.
- 62 Verify the formula for $\int \frac{1}{u\sqrt{u^2 - a^2}} du$.
- 63 As blood flows through a blood vessel there is a loss of energy due to friction. According to **Poiseuille's Law**, this energy loss E is given by $E = kl/r^4$, where k is a constant, and where r and l are the radius and length,

respectively, of the blood vessel. Suppose a blood vessel of radius r_2 and length l_2 branches off, at an angle θ , from a blood vessel of radius r_1 and length l_1 , as illustrated in the accompanying figure, where the colored arrows indicate the direction of blood flow. The energy loss is then the sum of the individual energy losses; that is,

$$E = (kl_1/r_1^4) + (kl_2/r_2^4).$$

Express l_1 and l_2 in terms of a , b , and θ , and find the angle that minimizes the energy loss.

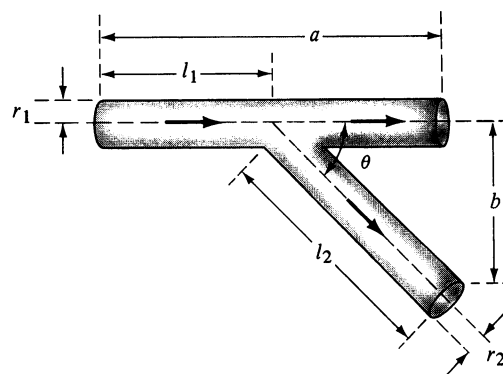


FIGURE FOR EXERCISE 63

8.5 The Hyperbolic Functions

The exponential expressions

$$\frac{e^x - e^{-x}}{2} \quad \text{and} \quad \frac{e^x + e^{-x}}{2}$$

occur frequently in applied mathematics and engineering. Their behavior is, in many ways, similar to that of $\sin x$ and $\cos x$. For reasons to be discussed later in this section, the expressions are called the *hyperbolic sine* and the *hyperbolic cosine* of x and are used to define the following **hyperbolic functions**.

Definition (8.10)

The **hyperbolic sine function**, denoted by **sinh**, and the **hyperbolic cosine function**, denoted **cosh**, are defined by

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

where x is any real number.

The graph of $y = \cosh x$ may be found by the method called **addition of ordinates**. To use this technique, the graphs of $y = \frac{1}{2}e^x$ and $y = \frac{1}{2}e^{-x}$ are sketched as indicated by the dashes in Figure 8.7. The y -coordinates of points on the graph of $y = \cosh x$ may then be obtained by adding y -coordinates of points on the other two graphs. This results in the sketch shown in the figure. It is evident from the graph that the range of $\cosh$ is $[1, \infty)$. This can also be proved directly.

The graph of $y = \sinh x$ may be obtained by adding y -coordinates of the graphs of $y = \frac{1}{2}e^x$ and $y = -\frac{1}{2}e^{-x}$ as illustrated in Figure 8.8.

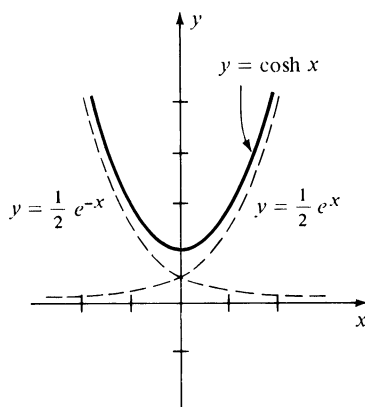


FIGURE 8.7

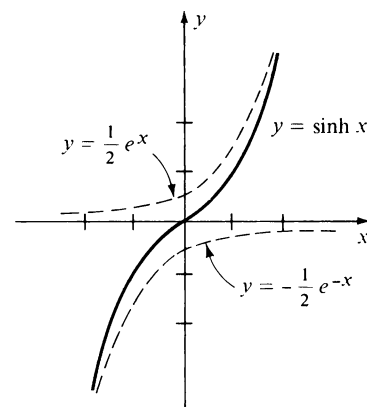
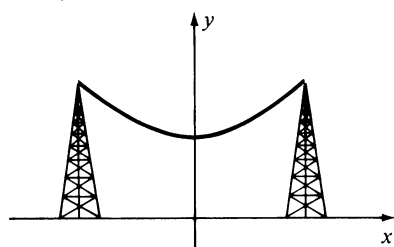


FIGURE 8.8

The hyperbolic cosine function can be used to describe the shape of a uniform flexible cable, or chain, whose ends are supported from the same height. This is often the case for telephone or power lines, as illustrated in Figure 8.9. At first glance the shape of the cable has the general appearance of a parabola, but this is not the case. If we introduce a coordinate system as indicated in the figure, then it can be shown that an equation that corresponds to the shape of the cable is $y = a \cosh (x/a)$, where a is a real number. The graph is called a **catenary**, after the Latin word for *chain*.

FIGURE 8.9 Catenary; $y = a \cosh \frac{x}{a}$

Another application of the hyperbolic cosine function occurs in the analysis of motion in a resisting medium. If an object is dropped from a given height, and if air resistance is neglected, then the distance y that it falls in t seconds is $y = \frac{1}{2}gt^2$, where g is a gravitational constant. However, air resistance cannot always be neglected. Indeed, as the velocity of the object increases, air resistance may significantly affect its motion. For example, if the air resistance is directly proportional to the square of the velocity, then it can be shown that the distance y that the object falls in t seconds is

$$y = A \ln (\cosh Bt)$$

where A and B are constants.

Many identities similar to those for trigonometric functions hold for the hyperbolic sine and cosine functions. For example, if $\cosh^2 x$ and $\sinh^2 x$ denote $(\cosh x)^2$ and $(\sinh x)^2$, respectively, we have the following.

Theorem (8.11)

$$\cosh^2 x - \sinh^2 x = 1$$

Proof By Definition (8.10),

$$\begin{aligned}
 \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\
 &= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4} \\
 &= \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4} \\
 &= \frac{4}{4} = 1.
 \end{aligned}$$

□

Theorem (8.11) is analogous to the identity $\cos^2 x + \sin^2 x = 1$. Many other hyperbolic identities may be established. Some will be found in the exercises. In each case it is sufficient to express the hyperbolic functions in terms of exponential functions and show that one side of the equation can be transformed into the other. The hyperbolic identities are similar to (but not always identical to) certain trigonometric identities. Any differences that occur usually involve signs of terms.

An interesting geometric relationship exists between the points $(\cos t, \sin t)$ and $(\cosh t, \sinh t)$, where t is any real number. Let us consider the graphs of $x^2 + y^2 = 1$ and $x^2 - y^2 = 1$ sketched in Figure 8.10. The graph in (i) of the figure is a unit circle with center at the origin. The graph in (ii) is called a *hyperbola*. (Hyperbolas and their properties will be discussed in detail in Chapter 12.) Note first that since $\cos^2 t + \sin^2 t = 1$, the point $P(\cos t, \sin t)$ is on the circle $x^2 + y^2 = 1$. Next, by Theorem (8.11), $\cosh^2 t - \sinh^2 t = 1$, and hence the point $P(\cosh t, \sinh t)$ is on the hyperbola $x^2 - y^2 = 1$. These are the reasons for referring to $\cos$ and $\sin$ as *circular* functions (see Section 1.6) and to $\cosh$ and $\sinh$ as *hyperbolic* functions.

There is another analogy between the graphs in Figure 8.10. If $0 < t < \pi/2$, then t is the radian measure of angle POB shown in (i) of the figure. By Theorem (2.20), the area A of the shaded circular sector is $A = (\frac{1}{2})(1)^2 t = t/2$ and hence $t = 2A$. It can be shown that if $P(\cosh t, \sinh t)$ is the point in (ii) of Figure 8.10, then $t = 2A$, where A is the area of the shaded *hyperbolic* sector (see Exercise 53).

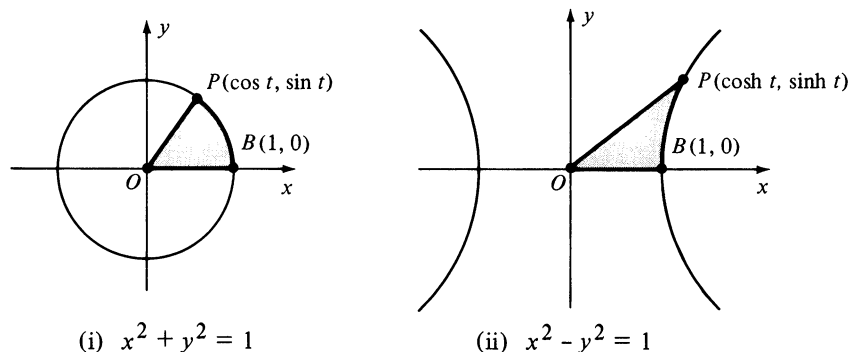


FIGURE 8.10

The striking analogies between the trigonometric and hyperbolic sine and cosine make it natural to introduce functions that correspond to the four remaining trigonometric functions. Specifically, the hyperbolic tangent, cotangent, secant, and cosecant functions, denoted by $\tanh$, $\coth$, sech , and csch , respectively, are defined as follows.

Definition (8.12)

$$\begin{aligned}\tanh x &= \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \\ \coth x &= \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}, \quad x \neq 0 \\ \operatorname{sech} x &= \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \\ \operatorname{csch} x &= \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}, \quad x \neq 0\end{aligned}$$

The graph of $y = \tanh x$ is sketched in Figure 8.11. The lines $y = 1$ and $y = -1$ are horizontal asymptotes. We leave the verification of this, and the sketching of the graphs of the remaining hyperbolic functions, as exercises.

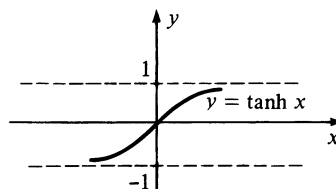


FIGURE 8.11

If we divide both sides of (8.11) by $\cosh^2 x$ and use the definitions of $\tanh x$ and $\operatorname{sech} x$ we obtain the first formula of the next theorem. The second formula may be obtained by dividing both sides of (8.11) by $\sinh^2 x$.

Theorem (8.13)

$$\begin{aligned}1 - \tanh^2 x &= \operatorname{sech}^2 x \\ \coth^2 x - 1 &= \operatorname{csch}^2 x\end{aligned}$$

Once again the reader should carefully note the differences between these and the analogous identities studied in trigonometry.

Derivative formulas for the hyperbolic functions are listed in the next theorem, where $u = g(x)$, and g is differentiable.

Theorem (8.14)

$D_x \sinh u = \cosh u D_x u$	$D_x \cosh u = \sinh u D_x u$
$D_x \tanh u = \operatorname{sech}^2 u D_x u$	$D_x \coth u = -\operatorname{csch}^2 u D_x u$
$D_x \operatorname{sech} u = -\operatorname{sech} u \tanh u D_x u$	$D_x \operatorname{csch} u = -\operatorname{csch} u \coth u D_x u$

Proof As usual, we consider only the case $u = x$. Since $D_x e^x = e^x$ and $D_x e^{-x} = -e^{-x}$,

$$D_x \sinh x = D_x \left(\frac{e^x - e^{-x}}{2} \right) = \frac{e^x + e^{-x}}{2} = \cosh x$$

$$D_x \cosh x = D_x \left(\frac{e^x + e^{-x}}{2} \right) = \frac{e^x - e^{-x}}{2} = \sinh x.$$

To differentiate $\tanh x$ we apply the Quotient Rule as follows.

$$\begin{aligned} D_x \tanh x &= D_x \frac{\sinh x}{\cosh x} \\ &= \frac{\cosh x D_x \sinh x - \sinh x D_x \cosh x}{\cosh^2 x} \\ &= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} \\ &= \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x \end{aligned}$$

The proofs of the remaining formulas are left as exercises. □

Example 1 Find $f'(x)$ if $f(x) = \cosh(x^2 + 1)$.

Solution Applying Theorem (8.14), with $u = x^2 + 1$,

$$\begin{aligned} f'(x) &= \sinh(x^2 + 1) \cdot D_x(x^2 + 1) \\ &= 2x \sinh(x^2 + 1). \end{aligned}$$
■

The integration formulas corresponding to the derivative formulas Theorem (8.14) are as follows.

Theorem (8.15)

$$\begin{array}{ll} \int \sinh u \, du = \cosh u + C & \int \cosh u \, du = \sinh u + C \\ \int \operatorname{sech}^2 u \, du = \tanh u + C & \int \operatorname{csch}^2 u \, du = -\coth u + C \\ \int \operatorname{sech} u \tanh u \, du = -\operatorname{sech} u + C & \int \operatorname{csch} u \coth u \, du = -\operatorname{csch} u + C \end{array}$$

Example 2 Evaluate $\int x^2 \sinh(x^3) \, dx$.

Solution If we let $u = x^3$, then $du = 3x^2 \, dx$ and

$$\begin{aligned} \int x^2 \sinh(x^3) \, dx &= \frac{1}{3} \int \sinh(x^3)(3x^2) \, dx \\ &= \frac{1}{3} \int \sinh u \, du = \frac{1}{3} \cosh u + C \\ &= \frac{1}{3} \cosh x^3 + C. \end{aligned}$$

8.5 Exercises

Verify the identities in Exercises 1–14.

1 $\cosh x + \sinh x = e^x$

2 $\cosh x - \sinh x = e^{-x}$

3 $\sinh(-x) = -\sinh x$

4 $\cosh(-x) = \cosh x$

5 $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$

6 $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$

7 $\sinh 2x = 2 \sinh x \cosh x$

8 $\cosh 2x = \cosh^2 x + \sinh^2 x$

9 $\tanh(x + y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$

10 $\tanh 2x = \frac{2 \tanh x}{1 + \tanh^2 x}$

11 $\cosh \frac{x}{2} = \sqrt{\frac{1 + \cosh x}{2}}$

12 $\tanh \frac{x}{2} = \frac{\sinh x}{1 + \cosh x}$

13 $\sinh x + \sinh y = 2 \sinh \frac{1}{2}(x + y) \cosh \frac{1}{2}(x - y)$

14 $(\cosh x + \sinh x)^n = \cosh nx + \sinh nx$ for every positive integer n (*Hint*: Use Exercise 1.)

In Exercises 15–30 find $f'(x)$ if $f(x)$ equals the given expression.

15 $\sinh 5x$

16 $\cosh \sqrt{4x^2 + 3}$

17 $\sqrt{x} \tanh \sqrt{x}$

18 $x \operatorname{csch} e^{4x}$

19 $(\operatorname{sech} x^2)/(x^2 + 1)$

20 $(\coth x)/(\cot x)$

- 21 $\cosh x^3$
 22 $\sinh(x^2 + 1)$
 23 $\cosh^3 x$
 24 $\sinh^2 3x$
 25 $\ln \sinh 2x$
 26 $\arctan \tanh x$
 27 $e^{3x} \operatorname{sech} x$
 28 $\sqrt{\operatorname{sech} 5x}$
 29 $1/(\tanh x + 1)$
 30 $(1 + \cosh x)/(1 - \cosh x)$
 31 Find y' if $\sinh xy = ye^x$.
 32 Find y' if $x^2 \tanh y = \ln y$.

Evaluate the integrals in Exercises 33–44.

- 33 $\int \frac{\sinh \sqrt{x}}{\sqrt{x}} dx$ 34 $\int \frac{\cosh \ln x}{x} dx$
 35 $\int \coth x dx$ 36 $\int \frac{1}{\cosh^2 3x} dx$
 37 $\int \sinh x \cosh x dx$ 38 $\int \operatorname{sech}^2 x \tanh x dx$
 39 $\int \tanh 3x \operatorname{sech} 3x dx$ 40 $\int \sinh x \sqrt{\cosh x} dx$

$$41 \int \tanh^2 3x \operatorname{sech}^2 3x dx \quad 42 \int \tanh x dx$$

$$43 \int \frac{\operatorname{sech}^2 x}{1 - 2 \tanh x} dx \quad 44 \int \frac{e^{\sinh x}}{\operatorname{sech} x} dx$$

- 45 Find the area of the region bounded by the graphs of $y = \sinh 3x$, $y = 0$, and $x = 1$.
 46 Find the arc length of the graph of $y = \cosh x$ from $x = 0$ to $x = 1$.
 47 Find the points on the graph of $y = \sinh x$ at which the tangent line has slope 2.
 48 The region bounded by the graphs of $y = \cosh x$, $x = -1$, $x = 1$, and $y = 0$ is revolved about the x -axis. Find the volume of the resulting solid.
 49 Verify the graph of $y = \tanh x$ in Figure 8.11.

Sketch the graphs of the equations in Exercises 50–52.

- 50 $y = \coth x$
 51 $y = \operatorname{sech} x$
 52 $y = \operatorname{csch} x$
 53 If A is the shaded region in (ii) of Figure 8.10, show that $t = 2A$.
 54 Sketch the graph of $x^2 - y^2 = 1$ and show that as t varies, the point $P(\cosh t, \sinh t)$ traces the part of the graph in quadrants I and IV.

8.6 The Inverse Hyperbolic Functions

The hyperbolic sine function is continuous and increasing for all x and hence, by Theorem (7.26), has a continuous, increasing inverse function, denoted by $\sinh^{-1}$. Since $\sinh x$ is defined in terms of e^x , we might expect that $\sinh^{-1}$ can be expressed in terms of the inverse, $\ln$, of the natural exponential function. The first formula of the next theorem shows that this is, indeed, the case.

Theorem (8.16)

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}), \quad x \geq 1$$

$$\tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x}, \quad |x| < 1$$

$$\operatorname{sech}^{-1} x = \ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right), \quad 0 < x \leq 1$$

Proof To prove the formula for $\sinh^{-1} x$ we begin by noting that

$$y = \sinh^{-1} x \quad \text{if and only if} \quad x = \sinh y.$$

The last equation can be used to find an explicit form for $\sinh^{-1} x$. Thus, if

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

then
$$e^y - 2x - e^{-y} = 0.$$

Multiplying both sides by e^y yields

$$e^{2y} - 2xe^y - 1 = 0$$

and, applying the quadratic formula,

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} \quad \text{or} \quad e^y = x \pm \sqrt{x^2 + 1}.$$

Since e^y is never negative, the minus sign must be discarded. Doing this and then taking the natural logarithm of both sides of the equation, we obtain

$$y = \ln(x + \sqrt{x^2 + 1}),$$

that is,
$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1}).$$

The formulas for the remaining inverse hyperbolic functions are obtained in similar fashion. As was the case with trigonometric functions, it is sometimes necessary to restrict the domain for an inverse function to exist. For example, if the domain of $\cosh$ is restricted to the set of nonnegative real numbers, then the resulting function is continuous and increasing and its inverse function $\cosh^{-1}$ is defined by

$$y = \cosh^{-1} x \quad \text{if and only if} \quad \cosh y = x, \quad y \geq 0.$$

Employing techniques similar to those used for $\sinh^{-1} x$ leads to the logarithmic formula for $\cosh^{-1} x$.

In like manner, we write

$$y = \tanh^{-1} x \quad \text{if and only if} \quad \tanh y = x.$$

Using Definition (8.12), the last equation may be written

$$\frac{e^y - e^{-y}}{e^y + e^{-y}} = x.$$

Solving for y gives us the logarithmic form for $\tanh^{-1} x$.

Finally, if the domain of sech is restricted to nonnegative numbers, there results a one-to-one function and we define

$$y = \operatorname{sech}^{-1} x \quad \text{if and only if} \quad \operatorname{sech} y = x, \quad y \geq 0.$$

Again introducing the exponential form leads to the fourth formula in the statement of the theorem. $\square$

In the next theorem $u = g(x)$, where g is differentiable and x is suitably restricted.

Theorem (8.17)

$$D_x \sinh^{-1} u = \frac{1}{\sqrt{u^2 + 1}} D_x u$$

$$D_x \cosh^{-1} u = \frac{1}{\sqrt{u^2 - 1}} D_x u, \quad u > 1$$

$$D_x \tanh^{-1} u = \frac{1}{1 - u^2} D_x u, \quad |u| < 1$$

$$D_x \operatorname{sech}^{-1} u = \frac{-1}{u\sqrt{1 - u^2}} D_x u, \quad 0 < u < 1$$

Proof

It follows from Theorem (8.16) that

$$\begin{aligned} D_x \sinh^{-1} x &= \frac{1}{x + \sqrt{x^2 + 1}} \left(1 + \frac{x}{\sqrt{x^2 + 1}} \right) \\ &= \frac{\sqrt{x^2 + 1} + x}{(x + \sqrt{x^2 + 1})\sqrt{x^2 + 1}} \\ &= \frac{1}{\sqrt{x^2 + 1}}. \end{aligned}$$

This formula can be extended to $D_x \sinh^{-1} u$ by applying the Chain Rule in the usual way. The proofs of the remaining formulas are left as exercises. $\square$

Example 1 Find y' if $y = \sinh^{-1}(\tan x)$.

Solution Using Theorem (8.17) with $u = \tan x$,

$$\begin{aligned} y' &= \frac{1}{\sqrt{\tan^2 x + 1}} D_x \tan x = \frac{1}{\sqrt{\sec^2 x}} \sec^2 x \\ &= \frac{1}{|\sec x|} |\sec x|^2 = |\sec x|. \end{aligned}$$

■

The following theorem may be verified by differentiating the right-hand side of each formula, where $u = g(x)$.

Theorem (8.18)

$$\int \frac{1}{\sqrt{u^2 + a^2}} du = \sinh^{-1} \frac{u}{a} + C, \quad a > 0$$

$$\int \frac{1}{\sqrt{u^2 - a^2}} du = \cosh^{-1} \frac{u}{a} + C, \quad u > a > 0$$

$$\int \frac{1}{a^2 - u^2} du = \frac{1}{a} \tanh^{-1} \frac{u}{a} + C, \quad a > 0, \quad |u| < a$$

$$\int \frac{1}{u\sqrt{a^2 - u^2}} du = -\frac{1}{a} \operatorname{sech}^{-1} \frac{|u|}{a} + C, \quad a > 0, \quad 0 < |u| < a$$

Example 2 Evaluate $\int \frac{1}{\sqrt{9x^2 + 25}} dx$.

Solution The integral may be expressed as in the first formula of Theorem (8.18) by letting $u = 3x$ and $du = 3 dx$. Thus,

$$\begin{aligned} \int \frac{1}{\sqrt{9x^2 + 25}} dx &= \frac{1}{3} \int \frac{1}{\sqrt{(3x)^2 + (5)^2}} 3 dx \\ &= \frac{1}{3} \sinh^{-1} \frac{3x}{5} + C. \end{aligned}$$

Example 3 Evaluate $\int \frac{e^x}{16 - e^{2x}} dx$.

Solution Letting $u = e^x$, $du = e^x dx$, and applying Theorem (8.18), we obtain

$$\begin{aligned} \int \frac{e^x}{16 - e^{2x}} dx &= \int \frac{1}{4^2 - (e^x)^2} e^x dx \\ &= \int \frac{1}{4^2 - u^2} du \\ &= \frac{1}{4} \tanh^{-1} \frac{u}{4} + C \\ &= \frac{1}{4} \tanh^{-1} \frac{e^x}{4} + C \end{aligned}$$

provided $e^x < 4$.

8.6 Exercises

1 Derive the formula for $\cosh^{-1} x$ in Theorem (8.16).

2 Derive the formula for $\tanh^{-1} x$ in Theorem (8.16).

3 Verify the formula for $D_x \cosh^{-1} u$.

4 Verify the formula for $D_x \tanh^{-1} u$.

5 Verify the formula for $\int \frac{1}{\sqrt{u^2 - a^2}} du$.

6 Verify the formula for $\int \frac{1}{a^2 - u^2} du$.

Sketch the graphs of the equations in Exercises 7–10.

7 $y = \sinh^{-1} x$

8 $y = \cosh^{-1} x$

9 $y = \tanh^{-1} x$

10 $y = \operatorname{sech}^{-1} x$

In Exercises 11–20 find $f'(x)$ if $f(x)$ equals the given expression.

11 $\sinh^{-1} 5x$

12 $\sinh^{-1} e^x$

13 $\cosh^{-1} \sqrt{x}$

14 $\sqrt{\cosh^{-1} x}$

15 $\tanh^{-1} (x^2 - 1)$

16 $\tanh^{-1} \sin 3x$

17 $x \sinh^{-1} (1/x)$

18 $1/\sinh^{-1} x^2$

19 $\ln \cosh^{-1} 4x$

20 $\cosh^{-1} \ln 4x$

Evaluate the integrals in Exercises 21–28.

21 $\int \frac{1}{\sqrt{9x^2 + 25}} dx$

22 $\int \frac{1}{\sqrt{16x^2 - 9}} dx$

23 $\int \frac{1}{49 - 4x^2} dx$

24 $\int \frac{\sin x}{\sqrt{1 + \cos^2 x}} dx$

25 $\int \frac{e^x}{\sqrt{e^{2x} - 16}} dx$

26 $\int \frac{2}{5 - 3x^2} dx$

27 $\int \frac{1}{x\sqrt{9 - x^4}} dx$

28 $\int \frac{1}{\sqrt{5 - e^{2x}}} dx$

29 Prove that if $a > 0$, then

$$\int \frac{1}{\sqrt{u^2 + a^2}} du = \ln(u + \sqrt{u^2 + a^2}) + C.$$

30 Prove that if $u > a > 0$, then

$$\int \frac{1}{\sqrt{u^2 - a^2}} du = \ln(u + \sqrt{u^2 - a^2}) + C.$$

8.7 Review: Concepts

Define or discuss each of the following.

1 Differentiation formulas for trigonometric functions

2 Integration formulas for trigonometric functions

3 The inverse trigonometric functions

4 Differentiation and integration formulas involving inverse trigonometric functions

5 Hyperbolic functions

6 Inverse hyperbolic functions

Review: Exercises

Find $f'(x)$ if $f(x)$ is defined as in Exercises 1–44.

1 $\cos \sqrt{3x^2 + x}$

2 $x^2 \cot 2x$

3 $(\sec x + \tan x)^5$

4 $\sqrt[3]{x^3 + \csc 6x}$

5 $x^2 \operatorname{arcsec} x^2$

6 $\tan^{-1} (\ln 3x)$

7 $\frac{(3x + 7)^4}{\sin^{-1} 5x}$

8 $\frac{\sin 8x}{4x^2 - x}$

9 $(\cos x)^{x+1}$

11 $\sqrt{2x^2 + \operatorname{sech} 4x}$

13 $\ln (\csc^3 2x)$

15 $\frac{1}{2x + \sec^2 x}$

17 $e^{\cos x} + (\cos x)^e$

10 $5^{\tan 2x}$

12 $x \sec^{-1} 4x$

14 $\log (\sec 2x)$

16 $\cot \frac{1}{x} + \frac{1}{\cot x}$

18 $\frac{\ln \sinh x}{x}$

- 19 $\cosh e^{-5x}$
 21 $\tanh^{-1}(\tanh \sqrt[3]{x})$
 23 $2^{\arctan 2x}$
 25 $\sin^3 e^{-2x}$
 27 $e^{-x^2} \cot x^2$
 29 $\frac{\csc x + 1}{\cot x + 1}$
 31 $\sin^{-1} \sqrt{1 - x^2}$
 33 $\tan(\sin 3x)$
 35 $(\tan x + \tan^{-1} x)^4$
 37 $\tan^{-1}(\tan^{-1} x)$
 39 $e^{-x} \sinh e^{-x}$
 41 $\frac{\sinh x}{\cosh x - \sinh x}$
 43 $\sinh^{-1} x^2$
- 20 $(\cos x)^{\cot x}$
 22 $\sec(\sec x)$
 24 $(1 + \arcsin 2x)^{\sqrt{2}}$
 26 $\ln \sin(\pi/3)$
 28 $\sec 5x \tan 5x$
 30 $\frac{1 - x^2}{\arccos x}$
 32 $\sqrt{\sin^{-1}(1 - x^2)}$
 34 $\sqrt{\sin \sqrt{x}}$
 36 $e^{4x} \sec^{-1} e^{4x}$
 38 $e^{x \cosh x}$
 40 $\ln \tanh(5x + 1)$
 42 $\frac{1}{x} \tanh \frac{1}{x}$
 44 $\cosh^{-1} \tan x$

Evaluate the integrals in Exercises 45–76.

- 45 $\int \sin(3 - 5x) dx$
 47 $\int \frac{\sec^2 \sqrt{x}}{\sqrt{x}} dx$
 49 $\int (\cot 9x + \csc 9x) dx$
 51 $\int e^x \tan e^x dx$
 53 $\int (\csc 3x + 1)^2 dx$
 55 $\int \frac{\sin 4x}{\tan 4x} dx$
 57 $\int \frac{x}{4 + 9x^2} dx$
 59 $\int \frac{e^{2x}}{\sqrt{1 - e^{2x}}} dx$
 61 $\int \frac{x}{\operatorname{sech} x^2} dx$
 63 $\int_{-1/2}^{1/2} \frac{1}{\sqrt{1 - x^2}} dx$
 65 $\int \sec^2 x (1 + \tan x)^2 dx$
- 46 $\int \csc \frac{x}{2} \cot \frac{x}{2} dx$
 48 $\int \frac{\sec(1/x)}{x^2} dx$
 50 $\int x \csc^2(3x^2 + 4) dx$
 52 $\int \cot 2x \csc 2x dx$
 54 $\int \frac{\sin x + 1}{\cos x} dx$
 56 $\int x \cot x^2 dx$
 58 $\int \frac{1}{4 + 9x^2} dx$
 60 $\int \frac{e^x}{\sqrt{1 - e^{2x}}} dx$
 62 $\int \frac{1}{x\sqrt{x^4 - 1}} dx$
 64 $\int_0^{\pi/2} \frac{\cos x}{1 + \sin^2 x} dx$
 66 $\int (1 + \cos^2 x) \sin x dx$

- 67 $\int \frac{\csc^2 x}{2 + \cot x} dx$
 69 $\int \frac{\sinh(\ln x)}{x} dx$
 71 $\int \frac{1}{\sqrt{9 - 4x^2}} dx$
 73 $\int \frac{1}{x\sqrt{9 - 4x^2}} dx$
 75 $\int \frac{x}{\sqrt{25x^2 + 36}} dx$
- 68 $\int (\sin x) e^{\cos x + 1} dx$
 70 $\int \operatorname{sech}^2(1 - 2x) dx$
 72 $\int \frac{x}{\sqrt{9 - 4x^2}} dx$
 74 $\int \frac{1}{x\sqrt{4x^2 - 9}} dx$
 76 $\int \frac{1}{\sqrt{25x^2 + 36}} dx$

- 77 Find the points on the graph of $y = \sin^{-1} 3x$ at which the tangent line is parallel to the line through $A(2, -3)$ and $B(4, 7)$.
 78 Find an equation of the tangent line to the graph of $y = \pi \tan(y/x) + x^2 - 16$ at the point $(4, \pi)$.
 79 Find the local extrema of $f(x) = 8 \sec x + \csc x$ on the interval $(0, \pi/2)$ and describe where $f(x)$ is increasing or decreasing on that interval.
 80 Find the points of inflection and discuss the concavity of the graph of $y = x \sin^{-1} x$.
 81 The region between the graphs of $y = \tan x$ and the x -axis, from $x = 0$ to $x = \pi/4$, is revolved about the x -axis. Find the volume of the resulting solid. (Hint: $\tan^2 x = \sec^2 x - 1$.)
 82 Find the area of the region bounded by the graphs of $y = x/(1 + x^4)$, $x = 1$, and $y = 0$.
 83 Use differentials to approximate $(0.98)^2 \tan^{-1}(0.98)$.
 84 Find the arclength of the graph of $y = \ln \tanh(x/2)$ from $x = 1$ to $x = 2$.
 85 A person on level ground, $\frac{1}{2}$ km from a point at which a balloon was released, observes its vertical ascent. If the balloon is rising at a constant rate of 2 m/sec, find the rate at which the angle of elevation of the observer's line of sight is changing at the instant the balloon is 100 m above the level of the observer's eyes.
 86 A square picture having sides 2 ft long is hung on a wall such that the base is 6 ft above the floor. A person whose eye level is 5 ft above the floor approaches the picture at a rate of 2 ft/sec. If θ is the angle between the line of sight and the top and bottom of the picture, find (a) the rate at which θ is changing when the person is 8 ft from the wall, and (b) the distance from the wall at which θ has its maximum value.

9 Additional Techniques and Applications of Integrations

The formulas for integrals listed below were obtained previously, where if $u = f(x)$, then $du = f'(x) dx$.

$$\int u^n du = \frac{1}{n+1} u^{n+1} + C, \quad n \neq -1 \qquad \int \frac{1}{u} du = \ln |u| + C$$

$$\int e^u du = e^u + C \qquad \int a^u du = \frac{1}{\ln a} a^u + C$$

$$\int \sin u du = -\cos u + C \qquad \int \cos u du = \sin u + C$$

$$\int \tan u du = \ln |\sec u| + C \qquad \int \cot u du = \ln |\sin u| + C$$

$$\int \sec u du = \ln |\sec u + \tan u| + C \qquad \int \csc u du = \ln |\csc u - \cot u| + C$$

$$\int \sec u \tan u du = \sec u + C \qquad \int \csc u \cot u du = -\csc u + C$$

$$\int \sec^2 u du = \tan u + C \qquad \int \csc^2 u du = -\cot u + C$$

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1} \frac{u}{a} + C \qquad \int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \sec^{-1} \frac{u}{a} + C$$

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \tan^{-1} \frac{u}{a} + C$$

In this chapter we shall discuss techniques that can be used to evaluate many other types of integrals. We also consider applications of the definite integral to the problems of finding moments and centers of mass of certain solids.

9.1 Integration by Parts

The following result is useful for simplifying certain types of integrals.

Integration by Parts Formula (9.1)

If $u = f(x)$ and $v = g(x)$, where f' and g' are continuous, then

$$\int u \, dv = uv - \int v \, du.$$

Proof By the Product Rule,

$$D_x[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

or, equivalently,

$$f(x)g'(x) = D_x[f(x)g(x)] - g(x)f'(x).$$

Integrating both sides of the previous equation gives us

$$\int f(x)g'(x) \, dx = \int D_x[f(x)g(x)] \, dx - \int g(x)f'(x) \, dx.$$

By Theorem (5.29) the first integral on the right side equals $f(x)g(x) + C$. Since we obtain another constant of integration from the second integral, it is unnecessary to include C in the formula; that is,

$$\int f(x)g'(x) \, dx = f(x)g(x) - \int g(x)f'(x) \, dx.$$

Since $du = f'(x) \, dx$ and $dv = g'(x) \, dx$, the preceding formula may be written as in (9.1). $\square$

When applying Formula (9.1) to a given integral, we begin by letting one part of the integrand correspond to dv . The expression chosen for dv must include the differential dx . After selecting dv , the remaining part of the integrand is designated by u . Since this process involves splitting the integrand into two parts, the use of (9.1) is referred to as **integrating by parts**. A proper choice for dv is crucial. A good rule of thumb is to choose the most complicated part of the integrand that can be readily integrated. The following examples illustrate this important method of integration.

Example 1 Find $\int xe^{2x} \, dx$.

Solution There are four possible choices for dv , namely dx , $x \, dx$, $e^{2x} \, dx$, or $xe^{2x} \, dx$. If we let $dv = e^{2x} \, dx$, then the remaining part of the integrand is u ; that is, $u = x$. To find v we integrate dv , obtaining $v = \frac{1}{2}e^{2x}$. Note that a

constant of integration is not added at this stage of the solution. (In Exercise 51 you are asked to prove that if a constant is added to v , the same result is obtained.). Since $u = x$ we see that $du = dx$. For ease of reference it is convenient to display these expressions as follows:

$$\begin{aligned} dv &= e^{2x} dx & u &= x \\ v &= \frac{1}{2}e^{2x} & du &= dx. \end{aligned}$$

Substituting these expressions in Formula (9.1), that is, *integrating by parts*, we obtain

$$\int x e^{2x} dx = x(\frac{1}{2}e^{2x}) - \int \frac{1}{2}e^{2x} dx.$$

The integral on the right side may be found by means of Theorem (7.21). This gives us

$$\int x e^{2x} dx = \frac{1}{2}x e^{2x} - \frac{1}{4}e^{2x} + C. \quad \blacksquare$$

It takes considerable practice to become proficient in making a suitable choice for dv . To illustrate, if we had chosen $dv = x dx$ in Example 1, then it would have been necessary to let $u = e^{2x}$, giving us

$$\begin{aligned} dv &= x dx & u &= e^{2x} \\ v &= \frac{1}{2}x^2 & du &= 2e^{2x} dx. \end{aligned}$$

Integrating by parts, we obtain

$$\int x e^{2x} dx = \frac{1}{2}x^2 e^{2x} - \int x^2 e^{2x} dx.$$

Since the exponent associated with x has increased, the integral on the right is more complicated than the given integral. This indicates an incorrect choice for dv .

Example 2 Evaluate $\int x \sec^2 x dx$.

Solution Since $\sec^2 x$ can be integrated readily, we let $dv = \sec^2 x dx$. The remaining part of the integrand is x and hence we must let $u = x$. Thus

$$\begin{aligned} dv &= \sec^2 x dx & u &= x \\ v &= \tan x & du &= dx \end{aligned}$$

and integration by parts gives us

$$\begin{aligned} \int x \sec^2 x dx &= x \tan x - \int \tan x dx \\ &= x \tan x - \ln |\sec x| + C. \end{aligned} \quad \blacksquare$$

We did not state a formula for $\int \ln x \, dx$ in Chapter 7. The reason for not doing so is that integration by parts is needed to find an antiderivative of the natural logarithmic function, as shown in the next example.

Example 3 Find $\int \ln x \, dx$.

Solution Let

$$\begin{aligned} dv &= dx & u &= \ln x \\ v &= x & du &= \frac{1}{x} dx. \end{aligned}$$

Integrating by parts, we obtain

$$\begin{aligned} \int \ln x \, dx &= x \ln x - \int x \left(\frac{1}{x} \right) dx \\ &= x \ln x - \int dx \\ &= x \ln x - x + C. \end{aligned}$$

■

Sometimes it is necessary to use integration by parts more than once in the same problem. This is illustrated in the next example.

Example 4 Find $\int x^2 e^{2x} \, dx$.

Solution Let

$$\begin{aligned} dv &= e^{2x} dx & u &= x^2 \\ v &= \frac{1}{2} e^{2x} & du &= 2x \, dx. \end{aligned}$$

Integrating by parts, we obtain

$$\begin{aligned} \int x^2 e^{2x} \, dx &= x^2 \left(\frac{1}{2} e^{2x} \right) - \int \left(\frac{1}{2} e^{2x} \right) 2x \, dx \\ &= \frac{1}{2} x^2 e^{2x} - \int x e^{2x} \, dx. \end{aligned}$$

To evaluate the integral on the right side of the last equation we must again integrate by parts. Proceeding exactly as in Example 1 leads to

$$\int x^2 e^{2x} \, dx = \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x} + C.$$

■

The following example illustrates another device for finding certain integrals by means of two applications of the integration by parts formula.

Example 5 Find $\int e^x \cos x \, dx$.

Solution Let

$$\begin{aligned} dv &= \cos x \, dx & u &= e^x \\ v &= \sin x & du &= e^x \, dx. \end{aligned}$$

Integrating by parts,

$$(a) \quad \int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx.$$

We next apply integration by parts to the integral on the right side of equation (a). Letting

$$\begin{aligned} dv &= \sin x \, dx & u &= e^x \\ v &= -\cos x & du &= e^x \, dx \end{aligned}$$

and integrating by parts leads to

$$(b) \quad \int e^x \sin x \, dx = -e^x \cos x + \int e^x \cos x \, dx.$$

If we now use equation (b) to substitute on the right side of equation (a) we obtain

$$\int e^x \cos x \, dx = e^x \sin x - \left[-e^x \cos x + \int e^x \cos x \, dx \right]$$

$$\text{or} \quad \int e^x \cos x \, dx = e^x \sin x + e^x \cos x - \int e^x \cos x \, dx.$$

Adding $\int e^x \cos x \, dx$ to both sides gives us

$$2 \int e^x \cos x \, dx = e^x(\sin x + \cos x).$$

Finally, dividing both sides by 2 and adding the constant of integration, we have

$$\int e^x \cos x \, dx = \frac{1}{2}e^x(\sin x + \cos x) + C. \quad \blacksquare$$

Some care must be taken when evaluating integrals of the type given in Example 5. To illustrate, suppose in the evaluation of the integral on the right in equation (a) of the solution we had used

$$\begin{aligned} dv &= e^x \, dx & u &= \sin x \\ v &= e^x & du &= \cos x \, dx. \end{aligned}$$

In this event integration by parts leads to

$$\int e^x \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx.$$

If we now substitute in (a), we obtain

$$\int e^x \cos x \, dx = e^x \sin x - \left[e^x \sin x - \int e^x \cos x \, dx \right]$$

which reduces to

$$\int e^x \cos x \, dx = \int e^x \cos x \, dx.$$

Although this is a true statement, it is not a solution to the problem! Incidentally, the integral in Example 5 *can* be evaluated by using $dv = e^x \, dx$ for *both* the first and second applications of the integration by parts formula.

Integration by parts may sometimes be employed to obtain **reduction formulas** for integrals. Such formulas can be used to write an integral involving powers of an expression in terms of integrals that involve lower powers of the expression.

Example 6 Find a reduction formula for $\int \sin^n x \, dx$.

Solution Let

$$\begin{aligned} dv &= \sin x \, dx & u &= \sin^{n-1} x \\ v &= -\cos x & du &= (n-1)\sin^{n-2} x \cos x \, dx. \end{aligned}$$

Integrating by parts,

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx.$$

Since $\cos^2 x = 1 - \sin^2 x$, we may write

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx.$$

Consequently,

$$\int \sin^n x \, dx + (n-1) \int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx.$$

The left side of the last equation reduces to $n \int \sin^n x \, dx$. Dividing both sides by n , we obtain

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{(n-1)}{n} \int \sin^{n-2} x \, dx. \quad \blacksquare$$

Example 7 Use the reduction formula in Example 6 to evaluate $\int \sin^4 x \, dx$.

Solution Using the formula with $n = 4$ gives us

$$\int \sin^4 x \, dx = -\frac{1}{4} \cos x \sin^3 x + \frac{3}{4} \int \sin^2 x \, dx.$$

Applying the reduction formula, with $n = 2$, to the integral on the right,

$$\begin{aligned} \int \sin^2 x \, dx &= -\frac{1}{2} \cos x \sin x + \frac{1}{2} \int dx \\ &= -\frac{1}{2} \cos x \sin x + \frac{1}{2}x + C_1. \end{aligned}$$

Consequently,

$$\int \sin^4 x \, dx = -\frac{1}{4} \cos x \sin^3 x - \frac{3}{8} \cos x \sin x + \frac{3}{8}x + C. \quad \blacksquare$$

It should be evident that by repeated applications of the formula in Example 6 we can find $\int \sin^n x \, dx$ for any positive integer n , because these reductions either end with $\int \sin x \, dx$ or $\int dx$, and each of these can be evaluated easily.

9.1 Exercises

Evaluate the integrals in Exercises 1–38.

- | | | | |
|--------------------------------|-------------------------------|--|--|
| 1 $\int x e^{-x} \, dx$ | 2 $\int x \sin x \, dx$ | 17 $\int e^{-x} \sin x \, dx$ | 18 $\int e^{3x} \cos 2x \, dx$ |
| 3 $\int x^2 e^{3x} \, dx$ | 4 $\int x^2 \sin 4x \, dx$ | 19 $\int \sin x \ln \cos x \, dx$ | 20 $\int_0^1 x^3 e^{-x^2} \, dx$ |
| 5 $\int x \cos 5x \, dx$ | 6 $\int x e^{-2x} \, dx$ | 21 $\int \sec^3 x \, dx$ | 22 $\int \csc^5 x \, dx$ |
| 7 $\int x \sec x \tan x \, dx$ | 8 $\int x \csc^2 3x \, dx$ | 23 $\int_0^1 \frac{x^3}{\sqrt{x^2 + 1}} \, dx$ | 24 $\int \sin \ln x \, dx$ |
| 9 $\int x^2 \cos x \, dx$ | 10 $\int x^3 e^{-x} \, dx$ | 25 $\int_0^{\pi/2} x \sin 2x \, dx$ | 26 $\int_{\pi/6}^{\pi/4} x \sec^2 x \, dx$ |
| 11 $\int \tan^{-1} x \, dx$ | 12 $\int \sin^{-1} x \, dx$ | 27 $\int x(2x + 3)^{99} \, dx$ | 28 $\int \frac{x^5}{\sqrt{1 - x^3}} \, dx$ |
| 13 $\int \sqrt{x} \ln x \, dx$ | 14 $\int x^2 \ln x \, dx$ | 29 $\int e^{4x} \sin 5x \, dx$ | 30 $\int x^3 \cos(x^2) \, dx$ |
| 15 $\int x \csc^2 x \, dx$ | 16 $\int x \tan^{-1} x \, dx$ | 31 $\int (\ln x)^2 \, dx$ | 32 $\int x 2^x \, dx$ |
| | | 33 $\int x^3 \sinh x \, dx$ | 34 $\int (x + 4) \cosh 4x \, dx$ |

35 $\int \cos \sqrt{x} \, dx$

36 $\int \cot^{-1} 3x \, dx$

37 $\int \cos^{-1} x \, dx$

38 $\int (x+1)^{10}(x+2) \, dx$

Use integration by parts to derive the reduction formulas in Exercises 39–42.

39 $\int x^m e^x \, dx = x^m e^x - m \int x^{m-1} e^x \, dx$

40 $\int x^m \sin x \, dx = -x^m \cos x + m \int x^{m-1} \cos x \, dx$

41 $\int (\ln x)^m \, dx = x(\ln x)^m - m \int (\ln x)^{m-1} \, dx$

42 $\int \sec^m x \, dx = \frac{\sec^{m-2} x \tan x}{m-1} + \frac{m-2}{m-1} \int \sec^{m-2} x \, dx,$

provided that $m \neq 1$.

43 Use Exercise 39 to evaluate $\int x^5 e^x \, dx$.

44 Use Exercise 41 to evaluate $\int (\ln x)^4 \, dx$.

45 If $f(x) = \sin \sqrt{x}$, find the area of the region under the graph of f from 0 to π^2 .

46 The region between the graph of $y = x\sqrt{\sin x}$ and the x -axis from $x = 0$ to $x = \pi/2$ is revolved about the x -axis. Find the volume of the resulting solid.

47 The region bounded by the graphs of $y = \ln x$, $y = 0$, and $x = e$ is revolved about the y -axis. Find the volume of the resulting solid.

48 Suppose the force $f(x)$ acting at the point with coordinate x on a coordinate line l is given by $f(x) = x^5 \sqrt{x^3 + 1}$. Find the work done in moving an object from $x = 0$ to $x = 1$.

49 The ends of a water trough have the shape of the region between the graph of $y = \sin x$ and the x -axis from $x = \pi$ to $x = 2\pi$. If the trough is full of water, find the total force on one end.

50 The velocity (at time t) of a point moving along a coordinate line is t/e^{2t} ft/sec. If the point is at the origin at $t = 0$, find its position at time t .

51 When applying the Integration by Parts Formula (9.1), show that if, after choosing dv , we use $v + C$ in place of v , the same result is obtained.

52 In Chapter 6 the argument given for finding volumes by means of cylindrical shells (see Definition (6.7)) was incomplete because it was not shown that the same result is obtained if the disc method is also applicable. Use integration by parts to prove that if f is differentiable and either $f'(x) > 0$ on $[a, b]$ or $f'(x) < 0$ on $[a, b]$, and if V is the volume of the solid obtained by revolving the region bounded by the graphs of f , $x = a$, and $x = b$ about the x -axis, then the same value of V is obtained using either the disc method or the shell method. (*Hint*: Let g be the inverse function of f and use integration by parts on $\int_a^b \pi[f(x)]^2 \, dx$.)

53 Criticize the following use of Formula (9.1). Given $\int (1/x) \, dx$, let $dv = dx$ and $u = 1/x$, so that $v = x$ and $du = (-1/x^2) \, dx$. Hence

$$\int \frac{1}{x} \, dx = \left(\frac{1}{x}\right)x - \int x\left(-\frac{1}{x^2}\right) \, dx$$

or
$$\int \frac{1}{x} \, dx = 1 + \int \frac{1}{x} \, dx.$$

Consequently, $0 = 1$.

54 Prove that the analogue of Formula (9.1) for definite integrals is

$$\int_a^b u \, dv = uv \Big|_a^b - \int_a^b v \, du.$$

9.2 Trigonometric Integrals

In Example 6 of Section 9.1 a reduction formula was obtained for $\int \sin^n x \, dx$. Integrals of this type may also be found without resorting to integration by parts. If n is an odd positive integer, we begin by writing

$$\int \sin^n x \, dx = \int \sin^{n-1} x \sin x \, dx.$$

Since the integer $n - 1$ is even, we may then use the trigonometric identity $\sin^2 x = 1 - \cos^2 x$ to obtain a form that is easy to integrate.

Example 1 Evaluate $\int \sin^5 x \, dx$.

Solution As in the preceding discussion we have

$$\begin{aligned}\int \sin^5 x \, dx &= \int \sin^4 x \sin x \, dx \\ &= \int (\sin^2 x)^2 \sin x \, dx \\ &= \int (1 - \cos^2 x)^2 \sin x \, dx \\ &= \int (1 - 2\cos^2 x + \cos^4 x) \sin x \, dx.\end{aligned}$$

We next employ the method of substitution, letting

$$u = \cos x \quad \text{and} \quad du = -\sin x \, dx.$$

Thus

$$\begin{aligned}\int \sin^5 x \, dx &= -\int (1 - 2\cos^2 x + \cos^4 x)(-\sin x) \, dx \\ &= -\int (1 - 2u^2 + u^4) \, du \\ &= -u + \frac{2}{3}u^3 - \frac{1}{5}u^5 + C \\ &= -\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C. \quad \blacksquare\end{aligned}$$

A similar technique may be employed for odd powers of $\cos x$. Specifically, we write

$$\int \cos^n x \, dx = \int \cos^{n-1} x \cos x \, dx$$

and use the fact that $\cos^2 x = 1 - \sin^2 x$ to obtain an integrable form.

If the integrand is $\sin^n x$ or $\cos^n x$ and n is *even*, then the half-angle formulas

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \text{or} \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

may be used to simplify the integrand.

Example 2 Evaluate $\int \cos^2 x \, dx$.

Solution Using a half-angle formula,

$$\begin{aligned}\int \cos^2 x \, dx &= \frac{1}{2} \int (1 + \cos 2x) \, dx \\ &= \frac{1}{2}x + \frac{1}{4}\sin 2x + C. \quad \blacksquare\end{aligned}$$

Example 3 Evaluate $\int \sin^4 x \, dx$.

$$\begin{aligned}
 \text{Solution} \quad \int \sin^4 x \, dx &= \int (\sin^2 x)^2 \, dx \\
 &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 \, dx \\
 &= \frac{1}{4} \int (1 - 2 \cos 2x + \cos^2 2x) \, dx.
 \end{aligned}$$

We apply a half-angle formula again and write

$$\cos^2 2x = \frac{1}{2}(1 + \cos 4x) = \frac{1}{2} + \frac{1}{2} \cos 4x.$$

Substituting in the last integral and simplifying gives us

$$\begin{aligned}
 \int \sin^4 x \, dx &= \frac{1}{4} \int \left(\frac{3}{2} - 2 \cos 2x + \frac{1}{2} \cos 4x \right) \, dx \\
 &= \frac{3}{8}x - \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + C. \quad \blacksquare
 \end{aligned}$$

Integrals involving only products of $\sin x$ and $\cos x$ may be evaluated using the following guidelines.

(9.2) Guidelines for evaluating integrals of the form $\int \sin^m x \cos^n x \, dx$

- 1 If both m and n are even integers, use half-angle formulas for $\sin^2 x$ and $\cos^2 x$ to reduce the exponents by one-half.
- 2 If n is an odd integer, write the integral as

$$\int \sin^m x \cos^n x \, dx = \int \sin^m x \cos^{n-1} x \cos x \, dx$$

and express $\cos^{n-1} x$ in terms of $\sin x$ by using the trigonometric identity $\cos^2 x = 1 - \sin^2 x$. The substitution $u = \sin x$ then leads to an integrand that can be handled easily.

- 3 If m is an odd integer, write the integral as

$$\int \sin^m x \cos^n x \, dx = \int \sin^{m-1} x \cos^n x \sin x \, dx$$

and express $\sin^{m-1} x$ in terms of $\cos x$ by using the trigonometric identity $\sin^2 x = 1 - \cos^2 x$. Use the substitution $u = \cos x$ to evaluate the resulting integral.

Example 4 Evaluate $\int \cos^3 x \sin^4 x \, dx$.

Solution Using Guideline 2 of (9.2),

$$\begin{aligned}\int \cos^3 x \sin^4 x \, dx &= \int \cos^2 x \sin^4 x \cos x \, dx \\ &= \int (1 - \sin^2 x) \sin^4 x \cos x \, dx.\end{aligned}$$

If we let $u = \sin x$, then $du = \cos x \, dx$ and the integral may be written

$$\begin{aligned}\int \cos^3 x \sin^4 x \, dx &= \int (1 - u^2)u^4 \, du = \int (u^4 - u^6) \, du \\ &= \frac{1}{5}u^5 - \frac{1}{7}u^7 + C \\ &= \frac{1}{5}\sin^5 x - \frac{1}{7}\sin^7 x + C. \quad \blacksquare\end{aligned}$$

(9.3) Guidelines for evaluating integrals of the form $\int \tan^m x \sec^n x \, dx$

1 If n is an even integer, write the integral as

$$\int \tan^m x \sec^{n-2} x \sec^2 x \, dx$$

and express $\sec^{n-2} x$ in terms of $\tan x$ by using the trigonometric identity $\sec^2 x = 1 + \tan^2 x$. The substitution $u = \tan x$ leads to a simple integral.

2 If m is an odd integer, write the integral as

$$\int \tan^{m-1} x \sec^{n-1} x \sec x \tan x \, dx.$$

Since $m - 1$ is even, $\tan^{m-1} x$ may be expressed in terms of $\sec x$ by means of the identity $\tan^2 x = \sec^2 x - 1$. The substitution $u = \sec x$ then leads to a form that is readily integrable.

3 If n is odd and m is even, then another method such as integration by parts should be used.

Example 5 Evaluate $\int \tan^2 x \sec^4 x \, dx$.

Solution Using Guideline 1 of (9.3),

$$\begin{aligned}\int \tan^2 x \sec^4 x \, dx &= \int \tan^2 x \sec^2 x \sec^2 x \, dx \\ &= \int \tan^2 x (\tan^2 x + 1) \sec^2 x \, dx.\end{aligned}$$

If we let $u = \tan x$, then $du = \sec^2 x \, dx$ and

$$\begin{aligned}\int \tan^2 x \sec^4 x \, dx &= \int u^2(u^2 + 1) \, du \\ &= \int (u^4 + u^2) \, du \\ &= \frac{1}{5}u^5 + \frac{1}{3}u^3 + C \\ &= \frac{1}{5}\tan^5 x + \frac{1}{3}\tan^3 x + C. \quad \blacksquare\end{aligned}$$

Example 6 Evaluate $\int \tan^3 x \sec^5 x \, dx$.

Solution Using Guideline 2 of (9.3),

$$\begin{aligned}\int \tan^3 x \sec^5 x \, dx &= \int \tan^2 x \sec^4 x (\sec x \tan x) \, dx \\ &= \int (\sec^2 x - 1) \sec^4 x (\sec x \tan x) \, dx.\end{aligned}$$

Substituting $u = \sec x$ and $du = \sec x \tan x \, dx$, we obtain

$$\begin{aligned}\int \tan^3 x \sec^5 x \, dx &= \int (u^2 - 1)u^4 \, du \\ &= \int (u^6 - u^4) \, du \\ &= \frac{1}{7}u^7 - \frac{1}{5}u^5 + C \\ &= \frac{1}{7}\sec^7 x - \frac{1}{5}\sec^5 x + C. \quad \blacksquare\end{aligned}$$

Integrals of the form $\int \cot^m x \csc^n x \, dx$ may be evaluated in similar fashion.

Finally, integrals of the form $\int \sin mx \cos nx \, dx$ may be evaluated by means of the product formulas (see (1.36)), as illustrated in the next example.

Example 7 Evaluate $\int \cos 5x \cos 3x \, dx$.

Solution We may write

$$\cos 5x \cos 3x = \frac{1}{2}(\cos 8x + \cos 2x).$$

Consequently,

$$\begin{aligned}\int \cos 5x \cos 3x \, dx &= \frac{1}{2} \int (\cos 8x + \cos 2x) \, dx \\ &= \frac{1}{16} \sin 8x + \frac{1}{4} \sin 2x + C.\end{aligned}$$

■

9.2 Exercises

Evaluate the integrals in Exercise 1–30.

- | | |
|---|--|
| 1 $\int \cos^3 x \, dx$ | 2 $\int \sin^2 2x \, dx$ |
| 3 $\int \sin^2 x \cos^2 x \, dx$ | 4 $\int \cos^7 x \, dx$ |
| 5 $\int \sin^3 x \cos^2 x \, dx$ | 6 $\int \sin^5 x \cos^3 x \, dx$ |
| 7 $\int \sin^6 x \, dx$ | 8 $\int \sin^4 x \cos^2 x \, dx$ |
| 9 $\int \tan^3 x \sec^4 x \, dx$ | 10 $\int \sec^6 x \, dx$ |
| 11 $\int \tan^3 x \sec^3 x \, dx$ | 12 $\int \tan^5 x \sec x \, dx$ |
| 13 $\int \tan^6 x \, dx$ | 14 $\int \cot^4 x \, dx$ |
| 15 $\int \sqrt{\sin x} \cos^3 x \, dx$ | 16 $\int \frac{\cos^3 x}{\sqrt{\sin x}} \, dx$ |
| 17 $\int (\tan x + \cot x)^2 \, dx$ | 18 $\int \cot^3 x \csc^3 x \, dx$ |
| 19 $\int_0^{\pi/4} \sin^3 x \, dx$ | 20 $\int_0^1 \tan^2 (\pi x/4) \, dx$ |
| 21 $\int \sin 5x \sin 3x \, dx$ | 22 $\int_0^{\pi/4} \cos x \cos 5x \, dx$ |
| 23 $\int_0^{\pi/2} \sin 3x \cos 2x \, dx$ | 24 $\int \sin 4x \cos 3x \, dx$ |
| 25 $\int \csc^4 x \cot^4 x \, dx$ | 26 $\int (1 + \sqrt{\cos x})^2 \sin x \, dx$ |
| 27 $\int \frac{\cos x}{2 - \sin x} \, dx$ | 28 $\int \frac{\tan^2 x - 1}{\sec^2 x} \, dx$ |
| 29 $\int \frac{\sec^2 x}{(1 + \tan x)^2} \, dx$ | 30 $\int \frac{\sec x}{\cot^5 x} \, dx$ |

- 31 The region bounded by the x -axis and the arc of $y = \cos^2 x$ from $x = 0$ to $x = 2\pi$ is revolved about the x -axis. Find the volume of the resulting solid.
- 32 The region between the graphs of $y = \tan^2 x$ and $y = 0$ from $x = 0$ to $x = \pi/4$ is revolved about the x -axis. Find the volume of the resulting solid.
- 33 Suppose the velocity (at time t) of a point moving on a coordinate line is $\cos^2 \pi t$ ft/sec. How far does the point travel in 5 seconds?
- 34 The acceleration (at time t) of a point moving along a coordinate line is $\sin^2 t \cos t$ ft/sec². At $t = 0$ the point is at the origin and its velocity is 10 ft/sec. Find its position at time t .
- 35 (a) Prove that if m and n are positive integers,

$$\begin{aligned}\int \sin mx \sin nx \, dx &= \begin{cases} \frac{\sin(m-n)x}{2(m-n)} - \frac{\sin(m+n)x}{2(m+n)} + C & \text{if } m \neq n \\ \frac{x}{2} - \frac{\sin 2mx}{4m} + C & \text{if } m = n \end{cases}\end{aligned}$$

- (b) Obtain formulas similar to that in part (a) for

$$\int \sin mx \cos nx \, dx \quad \text{and} \quad \int \cos mx \cos nx \, dx.$$

- 36 (a) Use (a) of Exercise 35 to prove that

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

- (b) Find

(i) $\int_{-\pi}^{\pi} \sin mx \cos nx \, dx$

(ii) $\int_{-\pi}^{\pi} \cos mx \cos nx \, dx$

9.3 Trigonometric Substitutions

If an integrand contains one of the expressions $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$, where $a > 0$, the radical sign can be eliminated by using the trigonometric substitution listed in the following table.

Given expression	Trigonometric substitution
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$

When making a trigonometric substitution we shall assume that θ is in the range of the corresponding inverse trigonometric function. Thus, for the substitution $x = a \sin \theta$ we have $-\pi/2 \leq \theta \leq \pi/2$. In this event, $\cos \theta \geq 0$ and

$$\begin{aligned}
 \sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} \\
 &= \sqrt{a^2(1 - \sin^2 \theta)} \\
 &= \sqrt{a^2 \cos^2 \theta} \\
 &= a \cos \theta.
 \end{aligned}$$

If $\sqrt{a^2 - x^2}$ occurs in a denominator, we add the restriction $|x| \neq a$, or equivalently, $-\pi/2 < \theta < \pi/2$.

Example 1 Evaluate $\int \frac{1}{x^2 \sqrt{16 - x^2}} dx$.

Solution Let $x = 4 \sin \theta$, where $-\pi/2 < \theta < \pi/2$. It follows that

$$\sqrt{16 - x^2} = \sqrt{16 - 16 \sin^2 \theta} = 4\sqrt{1 - \sin^2 \theta} = 4 \cos \theta.$$

Since $x = 4 \sin \theta$, we have $dx = 4 \cos \theta d\theta$. Substituting in the given integral,

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{16 - x^2}} dx &= \int \frac{1}{(16 \sin^2 \theta) 4 \cos \theta} 4 \cos \theta d\theta \\
 &= \frac{1}{16} \int \frac{1}{\sin^2 \theta} d\theta \\
 &= \frac{1}{16} \int \csc^2 \theta d\theta \\
 &= -\frac{1}{16} \cot \theta + C.
 \end{aligned}$$

It is now necessary to return to the original variable of integration x . Since $\theta = \arcsin(x/4)$, we could express the result as $-\frac{1}{16} \cot \arcsin(x/4)$.

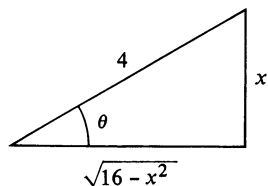


FIGURE 9.1 $\frac{x}{4} = \sin \theta$

However, since the given integral involves $\sqrt{16 - x^2}$, it is desirable to have the evaluated form also contain this radical. A method for accomplishing this is to use the following geometric device. If $0 < \theta < \pi/2$, then since $\sin \theta = x/4$, we may interpret θ as an acute angle of a right triangle having opposite side and hypotenuse of lengths x and 4 , respectively (see Figure 9.1). The length $\sqrt{16 - x^2}$ of the adjacent side is calculated by means of the Pythagorean Theorem. Referring to the triangle we see that

$$\cot \theta = \frac{\sqrt{16 - x^2}}{x}.$$

It can be shown that the last formula is also true if $-\pi/2 < \theta < 0$. Thus, Figure 9.1 may be used whether θ is positive or negative.

Substituting $\sqrt{16 - x^2}/x$ for $\cot \theta$ in our integral evaluation gives us

$$\begin{aligned} \int \frac{1}{x^2 \sqrt{16 - x^2}} dx &= -\frac{1}{16} \cdot \frac{\sqrt{16 - x^2}}{x} + C \\ &= -\frac{\sqrt{16 - x^2}}{16x} + C. \end{aligned}$$

If an integrand contains $\sqrt{a^2 + x^2}$, where $a > 0$, then the substitution $x = a \tan \theta$ will eliminate the radical sign. When using this substitution it will be assumed that θ is in the range of the inverse tangent function; that is, $-\pi/2 < \theta < \pi/2$. In this event $\sec \theta > 0$ and

$$\begin{aligned} \sqrt{a^2 + x^2} &= \sqrt{a^2 + a^2 \tan^2 \theta} \\ &= \sqrt{a^2(1 + \tan^2 \theta)} \\ &= \sqrt{a^2 \sec^2 \theta} \\ &= a \sec \theta. \end{aligned}$$

After making this substitution and evaluating the resulting trigonometric integral, it is necessary to return to the variable x . The preceding formulas show that

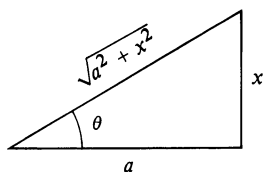


FIGURE 9.2 $\frac{x}{a} = \tan \theta$

$$\tan \theta = \frac{x}{a} \quad \text{and} \quad \sec \theta = \frac{\sqrt{a^2 + x^2}}{a}.$$

As in the solution of Example 1, the trigonometric functions of θ can be found by referring to the triangle in Figure 9.2, whether θ is positive or negative.

Example 2 Evaluate $\int \frac{1}{\sqrt{4 + x^2}} dx$.

Solution Let us substitute as follows:

$$x = 2 \tan \theta, \quad dx = 2 \sec^2 \theta d\theta.$$